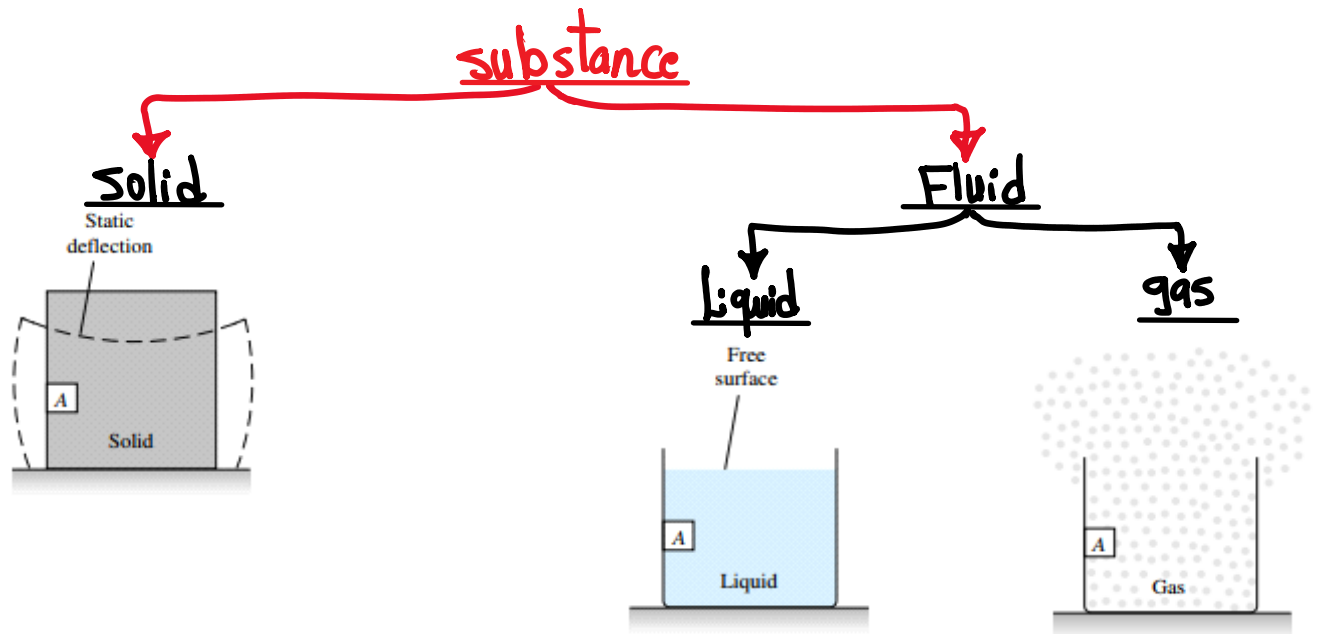
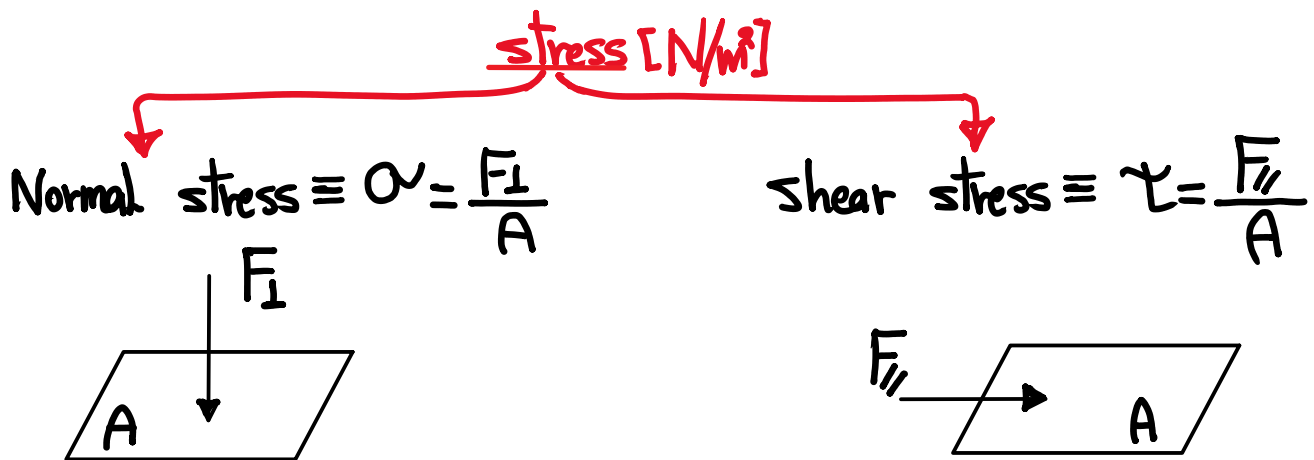


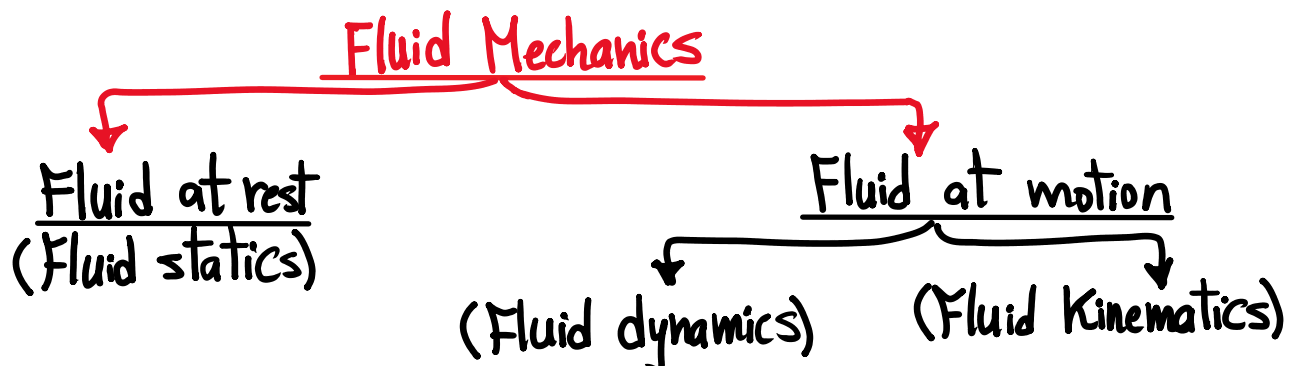
Chapter(1)



Fluid:- a substance, when acting with shear stress experienced shear strain rate (Continuous shear strain) $\equiv$ (Flow)



Fluid:- substance has no definite Volume, As the molecular attraction Forces $\equiv$ (Cohesive Forces) are very small (in liquid is smaller than solid) & (in gas is smaller than liquid)



Units & Dimensions

Units

to compare between quantities

basic unit systems

SI units

length $\equiv$ m

Mass $\equiv$ Kg

Time $\equiv$ Sec

Cgs units

length $\equiv$ cm

Mass $\equiv$ gm

Time $\equiv$ Sec

Dimensions

to identify the physical property

basic dimension systems

MLT system

M $\equiv$ mass

L $\equiv$ length

T $\equiv$ Time

FLT system

F $\equiv$ Force

L $\equiv$ length

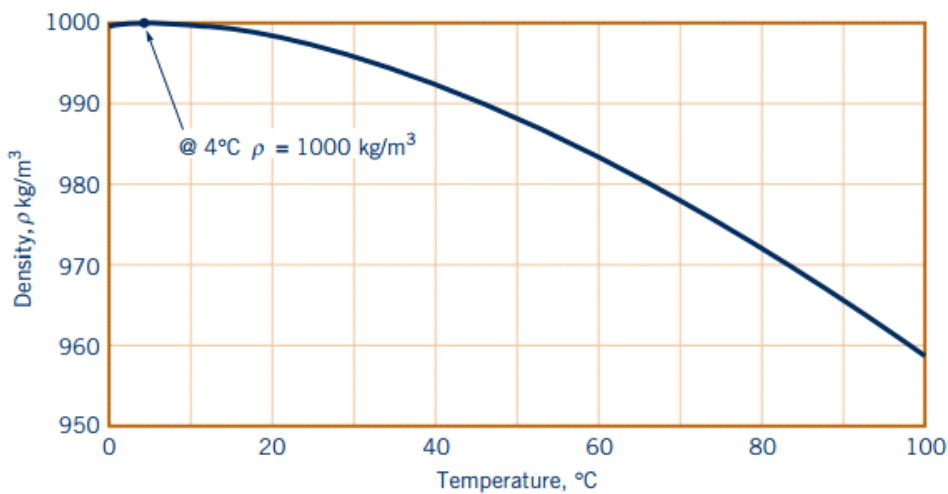
T $\equiv$ Time

Chapter (1): Fluid Properties

1- density $\equiv [\rho]$: mass per unit Volume, $\rho = \frac{m}{V}$ [Kg/m³] $\cdot$ [ML⁻³]

Important densities:-

$$\rho_{\text{std. liquid}} = \rho_w = \rho_{H_2O @ 4^\circ C} = 1000 \text{ Kg/m}^3$$



Density of water as a function of temperature.

$$\rho_{\text{std. gas}} = \rho_{\text{Air @ STC (1 bar, 298K)}} = \frac{P}{RT} = \frac{1 \times 10^5}{287 \times 298} = 1.17 \approx 1.2 \text{ Kg/m}^3$$

$$\rho_{\text{mercury}} = \rho_{Hg} = 13600 \text{ Kg/m}^3 \text{ (Heaviest Liquid)}$$

2-specific Volume $\equiv [\nu]$:- the Reciprocal of density, $\nu = \frac{1}{\rho}$

$$\therefore \nu = \frac{V}{m} \text{ [m}^3/\text{kg]}, \text{ [M}^{-1}\text{L}^3\text{]}$$

3-specific Weight $\equiv [\gamma]$:- Weight per unit Volume, $\gamma = \frac{W}{V}$

$$\therefore \gamma = \frac{mg}{V} = \rho g \text{ [N/m}^3\text{]} \equiv [\text{kg m/s}^2 \cdot \text{m}^3] \equiv [\text{kg/m}^2\text{s}^2], \text{ [ML}^{-2}\text{T}^{-2}\text{]}$$

4-specific gravity [S.G] $\equiv$ Relative density $[\rho_r]$:- [unitless], [Dimensionless]

For Liquid

$$S.G_{\text{liq}} = \frac{\rho_{\text{liq}}}{\rho_{\text{H}_2\text{O}}@4^\circ\text{C}}$$

$< 1000 \text{ kg/m}^3$

> 1 (sink with water) < 1 (float with water)

For Gas

$$S.G_{\text{gas}} = \frac{\rho_{\text{gas}}}{\rho_{\text{Air}}@STC}$$

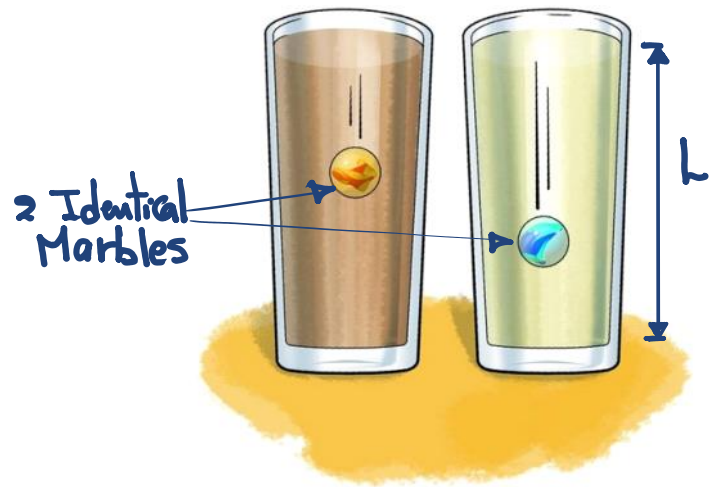
$< 1.2 \text{ kg/m}^3$

5 -Viscosity $\equiv$ absolute Viscosity $\equiv$ Coefficient of Viscosity $\equiv$ Dynamic Viscosity $\equiv [\mu]$

Fluid resistance to shear stress $\equiv [\text{Motion}] \equiv [\text{Flow}]$

How to experience the Viscosity :-

As the Fluid has Higher viscosity, the Marble will take more time to reach the bottom.



Newton's Experiment:-

$A \equiv$ Moving Plate Area $[\text{m}^2]$

$Y \equiv$ Gap between 2 plates $[\text{m}]$

$V \equiv$ Uniform velocity of plate $[\text{m/s}]$

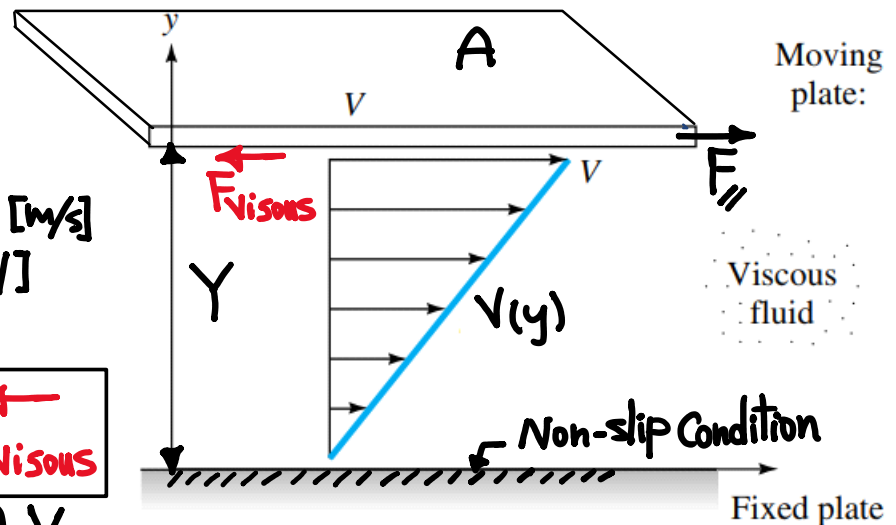
$F_{\parallel} \equiv$ Applied shear Force $[\text{N}]$

$$\vec{F}_{\parallel} \equiv \vec{F}_{\text{viscous}}$$

$$F_{\parallel} \propto \frac{AV}{Y}$$

$$\therefore \frac{F_{\parallel}}{A} \propto \frac{V}{Y}$$

$$\therefore \tau \propto \frac{V}{Y}$$



Newton's Law Viscosity:-

$$\therefore \tau = \mu \frac{V}{Y}$$

General newton's Law viscosity:-

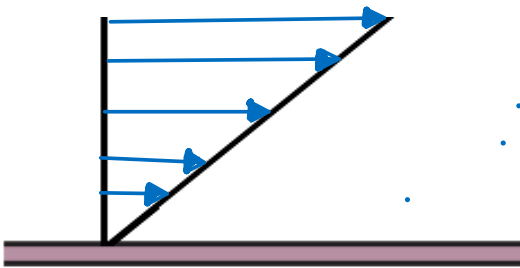
$$\therefore \tau = \mu \frac{dV}{dy}$$

$$\frac{dV}{dy} \text{ or } \frac{V}{Y} \text{ or } \frac{\Delta V}{\Delta y} \equiv [\text{Velocity Gradient}] \text{ or } [\text{shear strain rate}] \equiv \dot{\gamma} [1/\text{sec}]$$

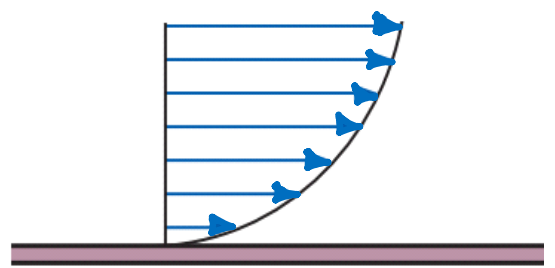
$$\therefore \tau = \mu \dot{\gamma}$$

Velocity Gradient $[\dot{\gamma}]$

Linear $(\frac{\Delta V}{\Delta y})$



non-Linear $(\frac{dV}{dy})$



Units of dynamic viscosity:-

$$\mu = \frac{\tau}{\dot{\gamma}}$$

SI units:-

$$[\text{Pa} \cdot \text{s}] \equiv [\frac{\text{N}}{\text{m}^2} \cdot \text{s}] \equiv [\frac{\text{kg m}}{\text{s}^2 \text{m}^2} \cdot \text{s}] \equiv [\text{kg/ms}], [M L^{-1} T^{-1}]$$

Cgs units:-

$$[\text{g/cm s}] \equiv [\text{Poise}]$$

$$1 \text{ Poise} = \frac{1}{10} \text{ Pa} \cdot \text{s}$$

$$1 \text{ centipoise} = \frac{1}{100} \text{ Poise} = \frac{1}{1000} \text{ Pa} \cdot \text{s}$$

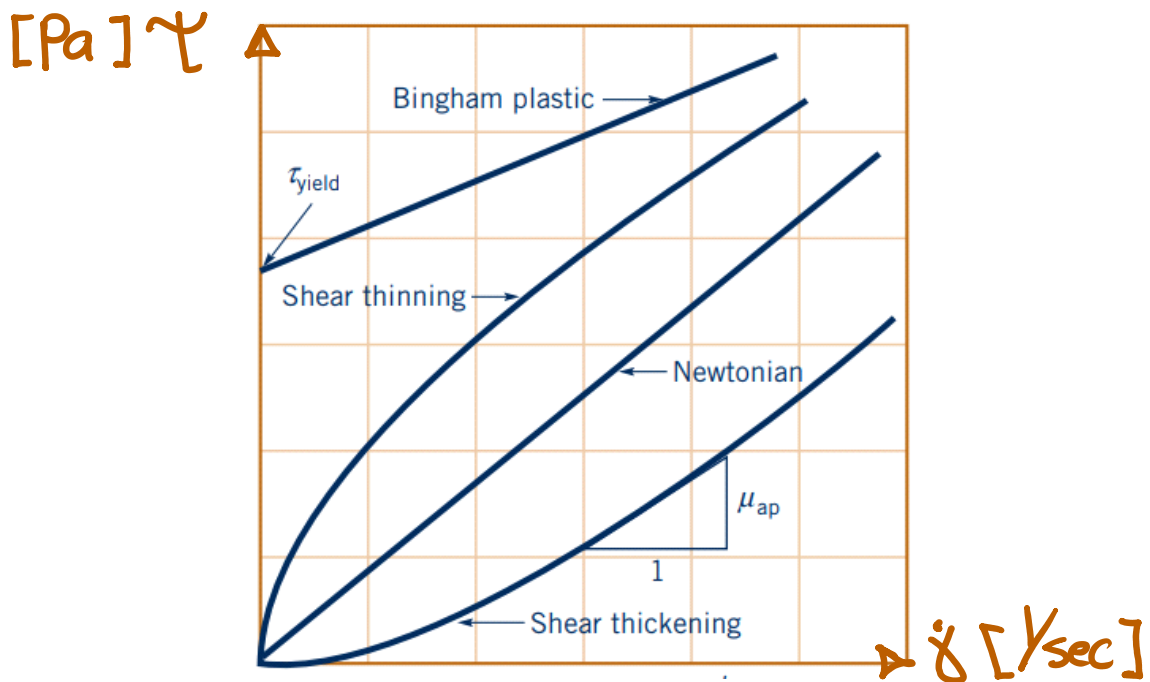
Fluid

Newtonian Fluid

- obeys Newton's law of viscosity
- τ & $\dot{\gamma}$ has linear relationship starting from origin
- $\mu \equiv \text{slope} \equiv \text{const.}$

Non-newtonian Fluid

- does not obey Newton's law of viscosity
- τ & $\dot{\gamma}$ has nonlinear relationship or linear but does not start from origin
- $\mu \equiv \text{apparent viscosity}$
 $\mu \neq \text{const}$



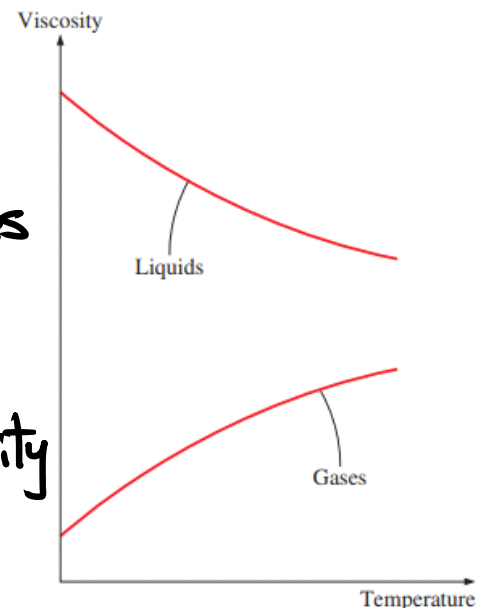
Effect of Temperature on Viscosity:-

Liquid Viscosity decreases As Temp. $\uparrow$

As liquid Viscosity depends on Cohesive Forces

Gas Viscosity increases As Temp. $\uparrow$

As gas viscosity depends on Molecular Activity

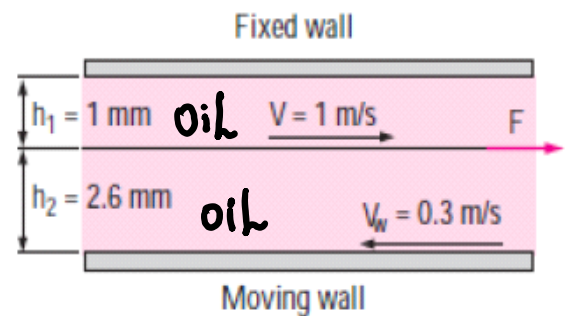


6-Kinematic Viscosity $\equiv$ Momentum diffusivity $\equiv [\nu]$:-

The ratio between dynamic Viscosity and density (Flow property)

$$\nu = \frac{\mu}{\rho} \equiv \left[\frac{\text{kg m}^3}{\text{m}^3 \text{s kg}} \right] \equiv [\text{m}^2/\text{s}], [L^2 T^{-1}]$$

16- A thin 20-cm x 20-cm flat plate is pulled at 1 m/s horizontally through a 3.6-mm-thick oil layer sandwiched between two plates, one stationary and the other moving at a constant velocity of 0.3 m/s. The dynamic viscosity of oil is 0.027 Pa.s. Assuming the velocity in each oil layer to vary linearly, (a) plot the velocity profile and find the location where the oil velocity is zero and (b) determine the force that needs to be applied on the plate to maintain this motion.



Given:- $A_{\text{ant}} = 20 \times 20 \text{ cm}^2$, $V_p = 1 \text{ m/s}$, $V_w = 0.3 \text{ m/s}$, $\mu = 0.027 \text{ Pa.s}$

Assuming Linear Velocity profile

Req:-

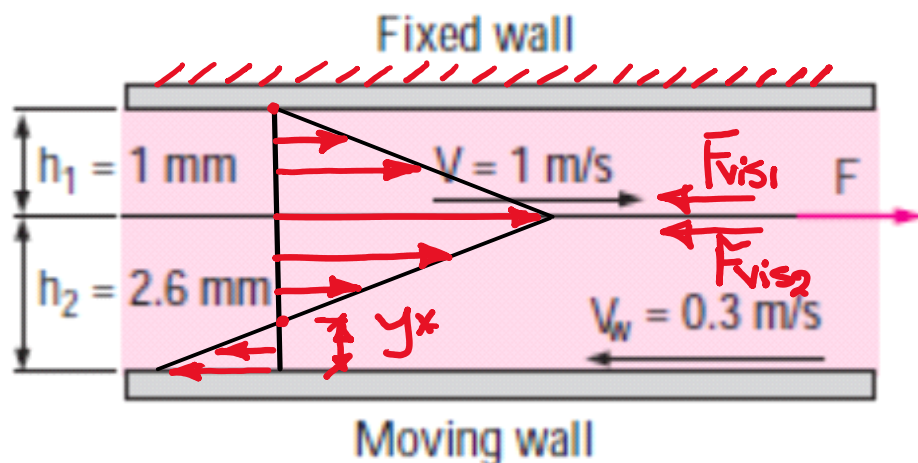
a) Velocity Profile

b) F

c) y^* (Where velocity = 0)

solution:-

a)



$$b) F = F_{vis1} + F_{vis2} = \tau_1 A + \tau_2 A = A[\tau_1 + \tau_2]$$

$$F = A \left[\mu \frac{\Delta V}{\Delta y} \right)_1 + \mu \frac{\Delta V}{\Delta y} \right)_2] = A \mu \left[\left(\frac{\Delta V}{\Delta y} \right)_1 + \left(\frac{\Delta V}{\Delta y} \right)_2 \right]$$

$$F = 20 \times 20 \times 10^{-4} \times 0.027 \left[\frac{1-0}{1 \times 10^{-3}} + \frac{1-(-0.3)}{2.6 \times 10^{-3}} \right]$$

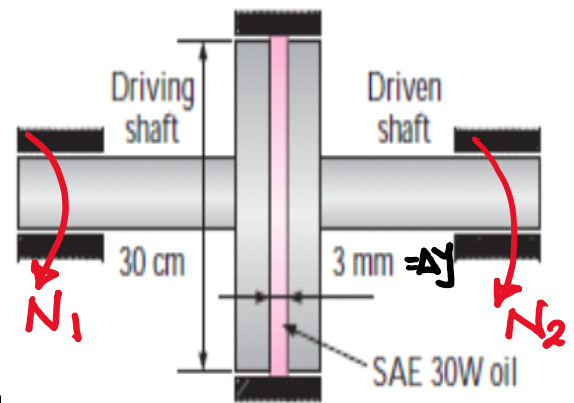
$$\boxed{F = 1.62 \text{ N}}$$

c) From similarity:-

$$\frac{1}{0.3} = \frac{2.6 - y^*}{y^*}$$

$$\boxed{y^* = 0.6 \text{ m}}$$

18- The clutch system shown is used to transmit torque through a 3-mm-thick oil film with $\mu = 0.38 \text{ N} \cdot \text{s/m}^2$ between two identical 30-cm-diameter disks. When the driving shaft rotates at a speed of 1450 rpm, the driven shaft is observed to rotate at 1398 rpm. Assuming a linear velocity profile for the oil film, determine the transmitted torque.

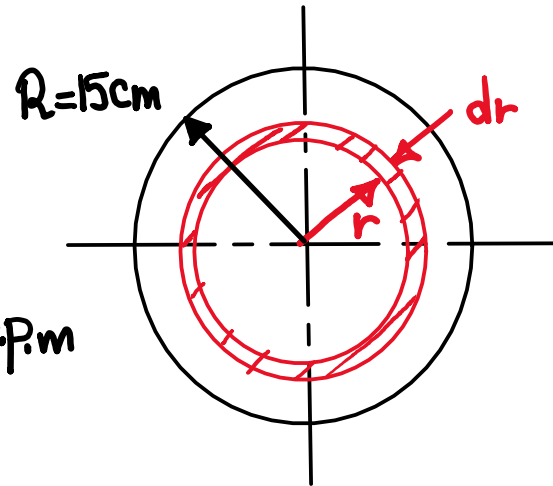
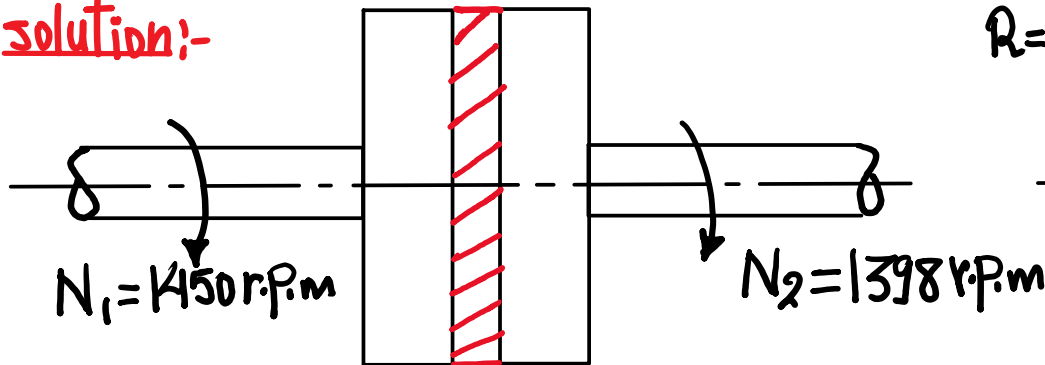


Given:- Clutch system, $\Delta y = 0.3 \text{ mm}$, $\mu = 0.38 \text{ Pa} \cdot \text{s}$

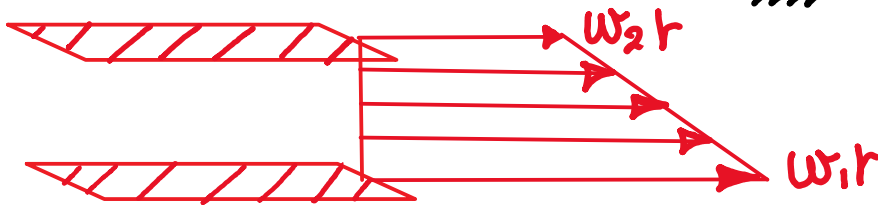
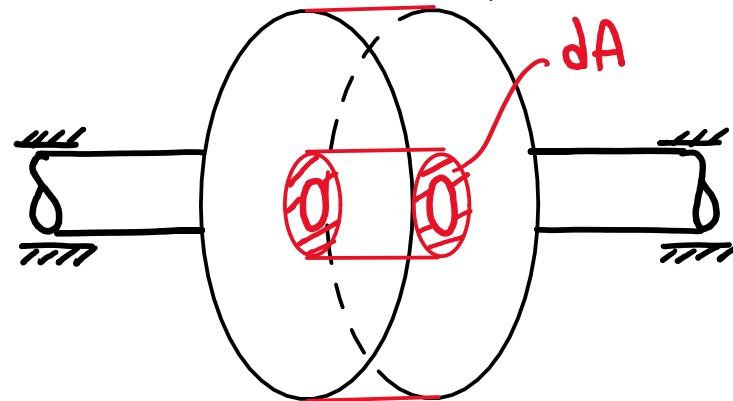
$N_1 = 1450 \text{ r.p.m}$ - $N_2 = 1398 \text{ r.p.m}$ - Assuming Linear Velocity Profile

Req:- T_{Trans}

solution:-



Linear Velocity Profile in Gap:-



For Elementary Area:-

$$dA = 2\pi r dr$$

Torque on Elementary Area:-

$$dT = dF r$$

$$dT = \gamma dA r = \int^M \frac{\Delta \gamma}{\Delta y} 2\pi r^2 dr = \int^M \frac{[\omega_1 r - \omega_2 r]}{\Delta y} 2\pi r^2 dr$$

$$dT = 2\pi \int^M \frac{[\omega_1 - \omega_2]}{\Delta y} r^3 dr$$

$$T = \int dT = \int_0^{R=0.15m} 2\pi \int^M \frac{[\omega_1 - \omega_2]}{\Delta y} r^3 dr$$

$$T = 2\pi \int^M \frac{[\omega_1 - \omega_2]}{\Delta y} \int_0^{R=0.15m} r^3 dr = 2\pi \int^M \frac{[\omega_1 - \omega_2]}{\Delta y} \frac{R^4}{4} \Big|_0^{0.15m}$$

$$T = 2\pi \times 0.38 \times \frac{\left[\frac{2\pi \times 1450}{60} - \frac{2\pi \times 1398}{60} \right]}{3 \times 10^{-3}} \times \frac{(0.15)^4}{4}$$

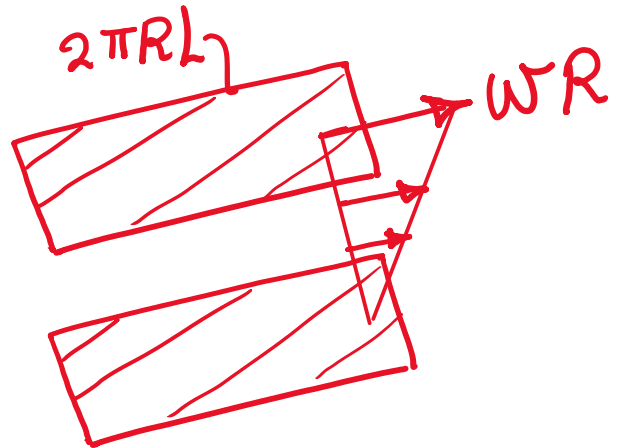
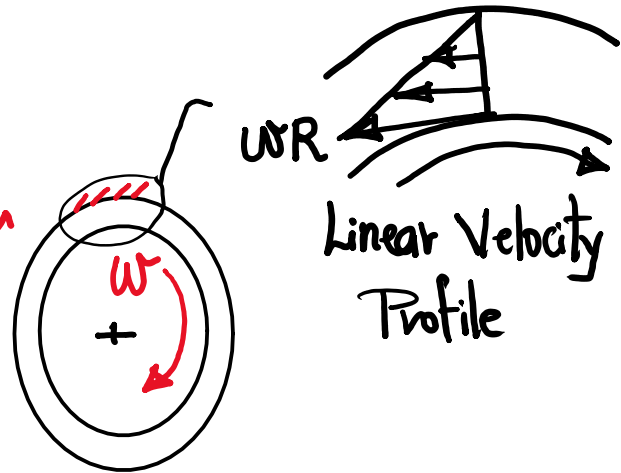
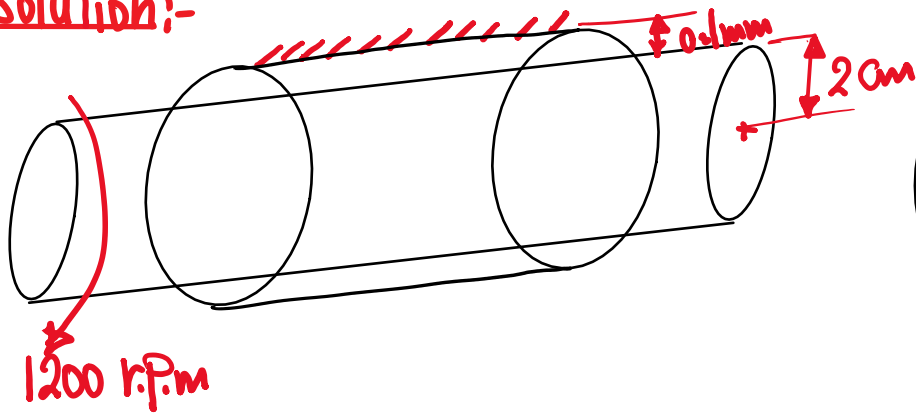
$$T = 0.832 \text{ N.m}$$

17-A lubricated shaft rotates inside a concentric sleeve bearing at 1200 rpm. The clearance δ is small compared with the radius R , so a linear velocity distribution in the lubricant may be assumed. What is the power required to rotate the shaft? $R = 2 \text{ cm}$, $L = 6 \text{ cm}$, $\delta = 0.1 \text{ mm}$ and $\mu = 0.2 \text{ N.s/m}^2$, where L is the bearing length. (Ans: 95.3 W)

Given:- A lubricated shaft, $N = 1200 \text{ r.p.m.}$, $\delta = 0.1 \text{ mm}$, $R = 2 \text{ cm}$
 $L = 6 \text{ cm}$, $\mu = 0.2 \text{ N.s/m}^2$

Req:- $\hat{P}_{\text{shaft}}$

Solution:-



$$\begin{aligned}\hat{P}_{\text{shaft}} &= T * \omega = F * R * \omega \\ &= \tau A R \omega \\ &= \mu \frac{\Delta v}{\Delta y} 2\pi R L R \omega\end{aligned}$$

$$= 2\pi \mu \frac{\omega R}{\delta} 2\pi R^2 \omega L$$

$$= 2\pi \mu \frac{\omega^2 R^3}{\delta} L = 2\pi * 0.2 * \frac{\left(\frac{2\pi * 1200}{60}\right)^2 * (2 * 10^{-2})^3}{0.1 * 10^{-3}} * 6 * 10^{-2}$$

$$\therefore \hat{P}_{\text{shaft}} = 95.25 \text{ W}$$

6. Bulk Modulus of Elasticity = [K] = [E_v]:-

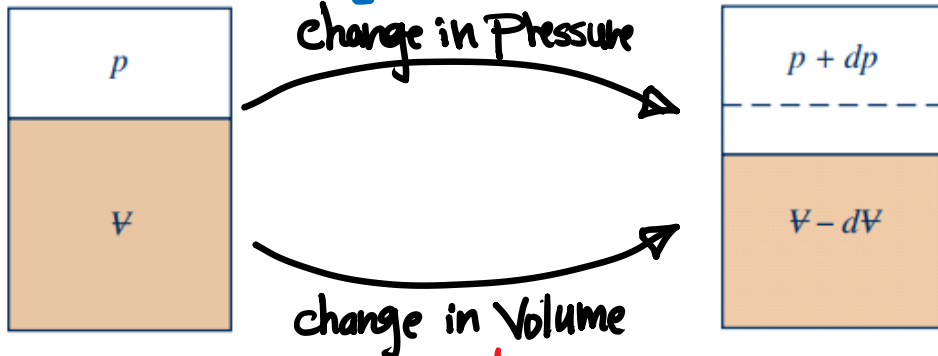
A property is commonly used to characterize fluid compressibility

$$E_v = \frac{dP}{-\frac{dV}{V}} \bigg|_{T=\text{const}} = \frac{dP}{\frac{d\rho}{\rho}} \bigg|_{T=\text{const}}, [P] = \left[\frac{N}{m^2}\right] = \frac{kg \cdot m}{s^2 \cdot m^2} = [kg/m \cdot s^2]$$

$$[M L^{-1} T^{-2}]$$

Initial Volume

Initial density



Fluid

Incompressible Fluid

$E_v \uparrow \uparrow \uparrow \uparrow$

$\therefore dV \downarrow \downarrow \downarrow \downarrow, d\rho \downarrow \downarrow \downarrow \downarrow$
 $V \approx \text{const}, \rho \approx \text{const}$

Ex: Liquids

Compressible Fluid

$E_v \downarrow \downarrow \downarrow \downarrow$

$\therefore dV \uparrow \uparrow \uparrow \uparrow, d\rho \uparrow \uparrow \uparrow \uparrow$
 $V \neq \text{const}, \rho \neq \text{const}$

Ex: Gases

(Note) For Gases: $E_v = P$

7. Volumetric Expansion Coefficient = β :-

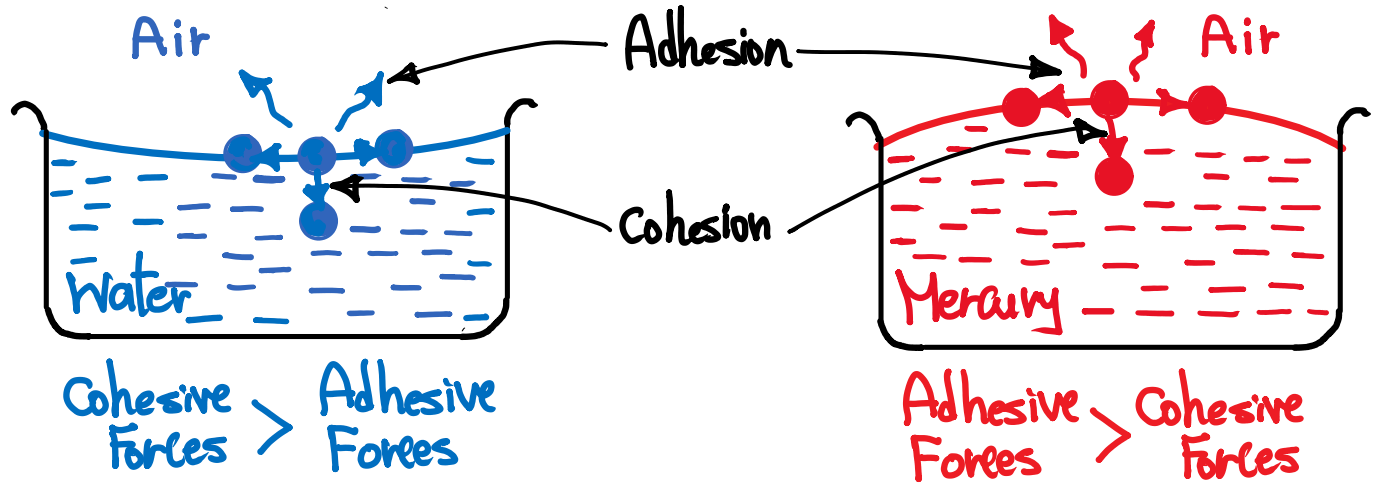
$$\beta = \frac{dV/V}{dT} \bigg|_{P=\text{const}}, [K^{-1}], [\theta^{-1}]$$

(Special Case) $\beta \equiv -\gamma_c$, for water beyond $4^\circ C$ ($A = \rho_w$ & V_w increased)

$$\beta_{\text{Ideal Gas}} = \frac{1}{T}, [T = K]$$

8. Surface tension $\equiv [\sigma]$:-

Net force Per unit length of liquid meniscus (free surface)



$$\sigma = \frac{F_{st}}{L}, [N/m] \equiv \left[\frac{kg \cdot m}{s^2 \cdot m} \right] \equiv [kg/s^2] = [MT^{-2}]$$

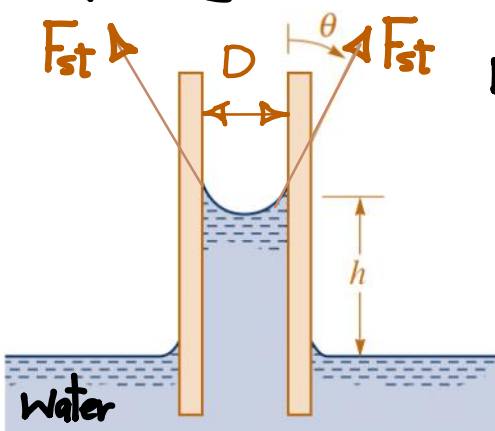
Applications On surface tension:-

A) Capillary Action

B) Droplet & Bubble

A) Capillary Action (For very small diameter)

Capillary rise

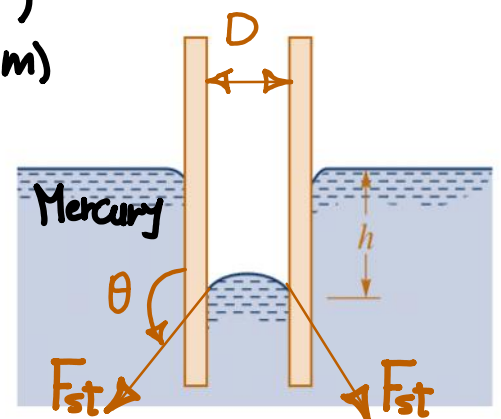


$h \equiv$ Capillary effect (m)

$\theta \equiv$ Contact angle ($^\circ$)

$D \equiv$ tube diameter (m)

Capillary drop



Adhesive Forces < Cohesive Forces

$\theta > 90^\circ$

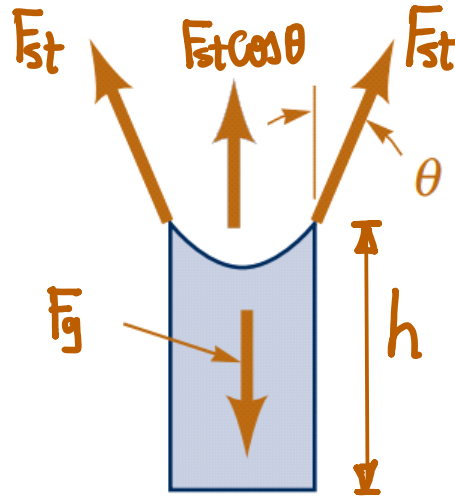
$h \equiv -ve$

Adhesive Forces > Cohesive Forces

$\theta < 90^\circ$

$h \equiv +ve$

How to get (h):-



For Force equilibrium:-

$$F_g = F_{st} \cos \theta$$

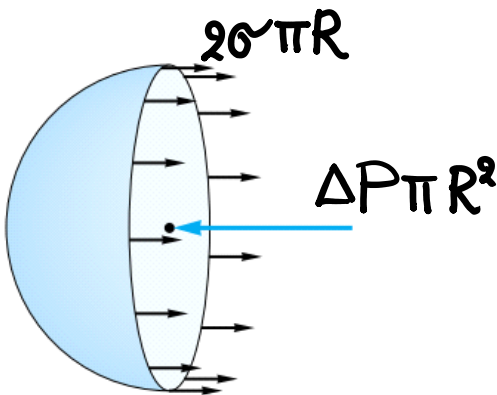
$$mg = 0 \pi D \cos \theta$$

$$\rho V g = 0 \pi D \cos \theta$$

$$\gamma \frac{\pi}{4} D^2 h = 0 \pi D \cos \theta$$

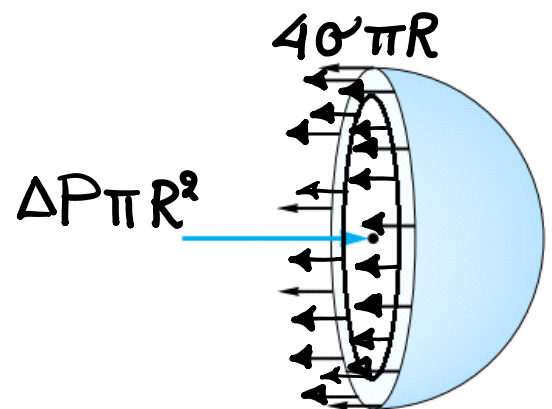
$$h = \frac{4\sigma \cos \theta}{\gamma D}$$

B) Droplet & Bubble



$$\Delta P \pi R^2 = 2\sigma \pi R$$

$$\Delta P = \frac{2\sigma}{R}$$

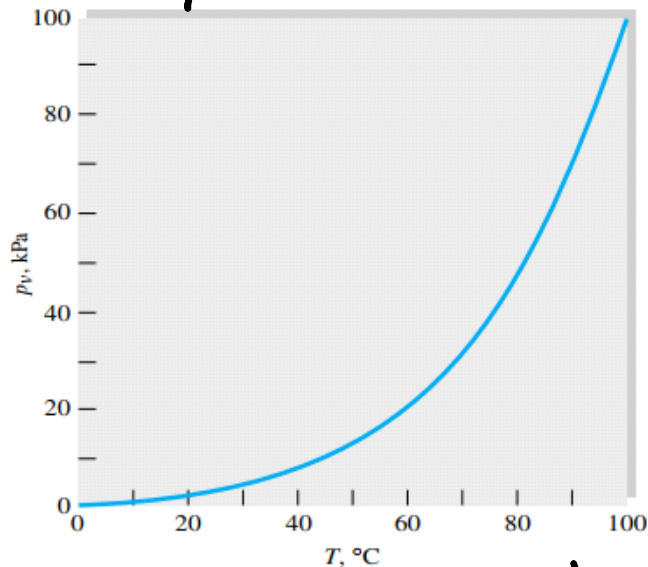


$$\Delta P \pi R^2 = 4\sigma \pi R$$

$$\Delta P = \frac{4\sigma}{R}$$

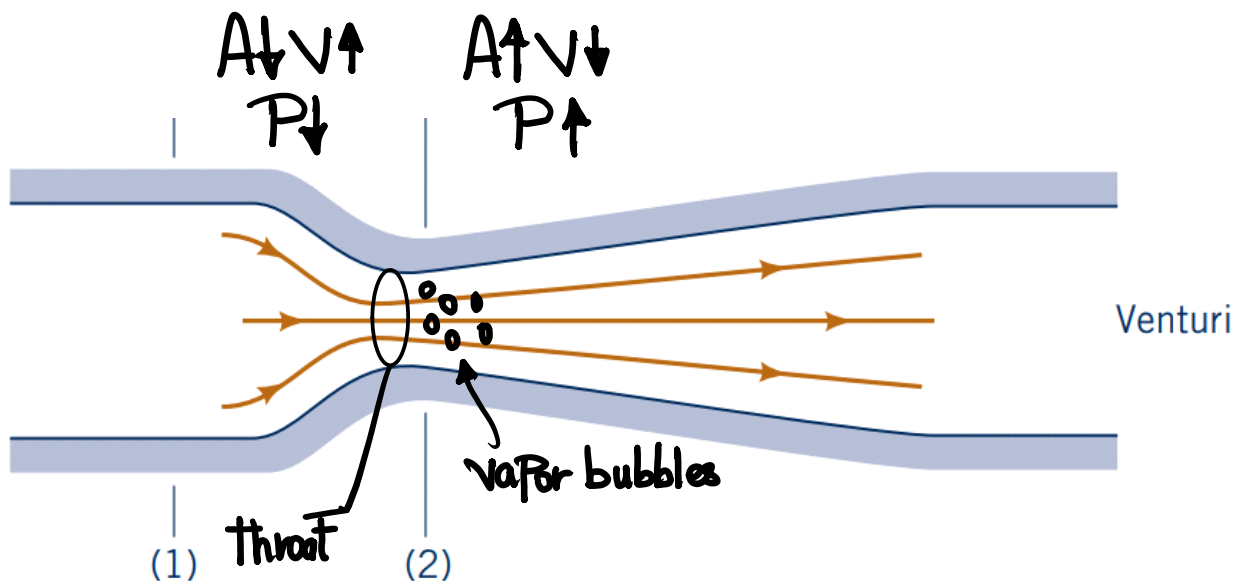
9. Vapor Pressure $\equiv [P_{vap}]$:- [KPa]

it is the Pressure at which a liquid boils and is in equilibrium with its own Vapor



Vapor Pressure of Water

Application :- [Cavitation]



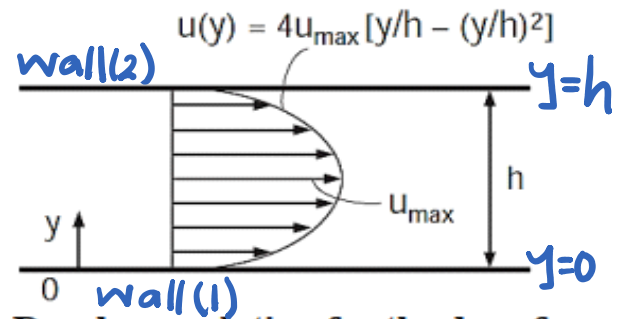
P_{min} occurs at min. diameter D_2

if $P_{min} < P_{vap}$ (Cavitation occurs)

Midterm: Fall 2021

Question (1): (8 points)

Consider laminar flow of a Newtonian fluid of viscosity μ between two parallel plates. The flow is one-dimensional, and the velocity profile is given as $u(y) = 4u_{\max} [y/h - (y/h)^2]$, where y is the vertical coordinate from the bottom surface, h is the distance between the two plates, and $u_{\max}$ is the maximum flow velocity that occurs at midplane. Develop a relation for the drag force exerted on both plates by the fluid in the flow direction per unit area of the plates. Develop a relation for the drag force exerted on both plates by the fluid in the flow direction per unit area of the plates.



Solution:-

$$u(y) = 4u_{\max} \left[\frac{y}{h} - \frac{y^2}{h^2} \right]$$

$$\therefore \tau(y) = \mu \frac{du}{dy}$$

$$\therefore \frac{du}{dy} = 4u_{\max} \left[\frac{1}{h} - 2 \frac{y}{h^2} \right]$$

$$\therefore \tau(y) = 4\mu u_{\max} \left[\frac{1}{h} - \frac{2y}{h^2} \right]$$

$$\text{So, } F_D = \tau(y) A = \tau(y) = 4\mu u_{\max} \left[\frac{1}{h} - \frac{2y}{h^2} \right]$$

For Wall (1):

$$\therefore \left. F_D \right|_{\text{at } y=0} = 4\mu u_{\max} \left[\frac{1}{h} \right]$$

For Wall (2):

$$\begin{aligned} \therefore \left. F_D \right|_{\text{at } y=h} &= 4\mu u_{\max} \left[\frac{1}{h} - \frac{2h}{h^2} \right] \\ &= -4\mu u_{\max} \left[\frac{1}{h} \right] \end{aligned}$$

The 2 Forces are equal but opposite in direction



Sheet (1)

Properties of Fluids

1-Write the dimensional representation and the SI units of the following:

- | | | | |
|----------------------|------------------------|--------------------|--------------|
| a) Density | b) Specific Weight | c) Pressure | d) Energy |
| e) γh | f) Discharge | g) Power | h) ρQv |
| i) Dynamic Viscosity | j) Kinematic Viscosity | k) Surface Tension | l) $V^2/2g$ |

2-If the specific weight of a gas is 12.40 N/m^3 , what is its specific volume in m^3/kg ?

3- Define:

- a) internal, external, and open-channel flows.
- b) incompressible flow and incompressible fluid. Must the flow of a compressible fluid necessarily be treated as compressible?
- c) What is the no-slip condition? What causes it?
- d) Define stress, normal stress, shear stress, and pressure.
- e) Can the coefficient of compressibility of a fluid be negative? How about the coefficient of volume expansion?
- f) Consider two identical small glass balls dropped into two identical containers, one filled with water and the other with oil. Which ball will reach the bottom of the container first? Why?

4-Determine the mass and the weight of the air contained in a room whose dimensions are 6 m x 6 m x 8 m. Assume the density of the air is 1.16 kg/m^3 .

5-Determine the specific weight, density and specific gravity of a fluid that occupies volume of 200 lit, and weighs 178 kg. Will this fluid float on the surface of an oil of specific gravity (0.80)? Provide Results in S.I. units.

6-The specific gravity of ethyl alcohol is 0.79. Calculate its specific weight and its mass density. Also, The kinematic viscosity of an oil, if the density of the oil is 800 kg/m^3 and its dynamic viscosity is $6 \times 10^{-3} \text{ kg/m.s}$

7-Water at 15°C and 1 atm pressure is heated to 100°C at constant pressure. Take the coefficient of volume expansion as $0.484 \times 10^{-3} \text{ K}^{-1}$, determine the change in the density of water. Consider the density of water at 1 atm is 999 kg/m^3 .

8-A water tank is completely filled with liquid water at 20°C . The tank material is such that it can withstand tension caused by a volume expansion of 2 percent.

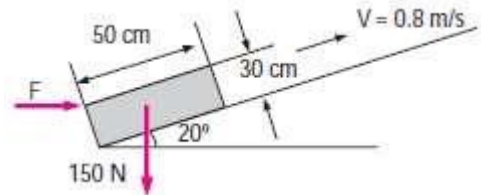
Determine the maximum temperature rise allowed without jeopardizing safety. Take the coefficient of volume expansion as $0.377 \times 10^{-3} \text{ K}^{-1}$.

9-In a fluid the velocity measured at a distance of 75mm from the boundary is 1.125m/s. The fluid has absolute viscosity 0.048 Pa s and relative density 0.913. What is the velocity gradient and shear stress at the boundary assuming a linear velocity distribution?

10-Calculate the capillary effect in millimeters in a glass tube of 4 mm diameter., when immersed in (i) water and (ii) in mercury. The temperature of liquid is 20° C and the values of surface tension of water and mercury at this temperature in contact with air are 0.0075 kg/m and 0.052 kg/m. The contact angle for water = 0 and for mercury = 130°.

11-Determine the force required to pull a 1.0 mm thick plate between two stationary parallel plates separated by a gap 4 mm. the moving plate is $(2 \times 3) \text{ m}^2$ surface area, and is located 0.5 mm from one of stationary plates. The moving plate has a velocity of 1.2 m/s and the viscosity of the lubricant is 0.07 poise.

12- A 50-cm x 30-cm x 20-cm block weighing 150 N is to be moved at a constant velocity of 0.8 m/s on an inclined surface with a friction coefficient of 0.27. (a) Determine the force F that needs to be applied in the horizontal direction. (b) If a 0.4-mm-thick oil film with a dynamic viscosity of 0.012 Pa.s is applied between the block and inclined surface, determine the percent reduction in the required force.



13-At the ocean surface under normal atmospheric pressure, the density of sea water is 1047 kg/m^3 . The absolute pressure is about 35 bar at the deepest point in the Pacific ocean where the depth is about 10860 m. What is the percentage change in the density of the sea water between the surface and the ocean bottom? Bulk modulus of elasticity of sea water = 2338.67 MPa.

(Ans: 0.15%)

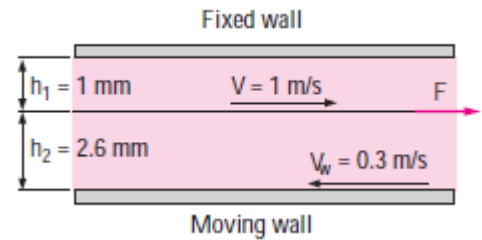
14-A 10-kg block slides down a smooth inclined surface, 20° with horizontal. Determine the terminal velocity of the block if the 0.1mm gap between the block and the surface contains oil of viscosity 3.8 cP (centipoises). Assume velocity distribution in the gap is linear and the area of block in contact with the oil is 0.1 m^2 .

(Ans: 8.83 m/s)

15-A large movable plate is located between two fixed plates at a distance 3.0 mm from one plate and 6.0 mm from the other. The spaces between the movable plate and the fixed plates are filled with oil of viscosity 0.02 and 0.01 Pa.s respectively. Determine the force required to pull the plates at 4 m/s if the surface area of the plate is 0.9 m^2 and its thickness is negligible. If one oil is used to fill the spaces between the plates ($\mu = 0.01 \text{ Pa.s}$), what is the minimum force required to pull the plate at the same velocity? Determine the location of the movable plate with respect to the fixed plates in this case.

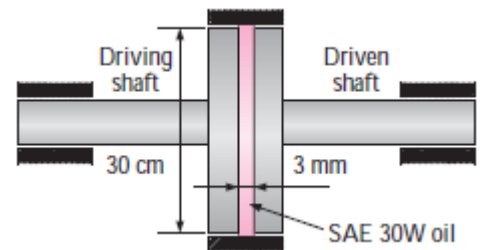
(Ans: 29.99 N, 16 N, the plate will be in the middle)

16- A thin 20-cm x 20-cm flat plate is pulled at 1 m/s horizontally through a 3.6-mm-thick oil layer sandwiched between two plates, one stationary and the other moving at a constant velocity of 0.3 m/s. The dynamic viscosity of oil is 0.027 Pa.s. Assuming the velocity in each oil layer to vary linearly, (a) plot the velocity profile and find the location where the oil velocity is zero and (b) determine the force that needs to be applied on the plate to maintain this motion.

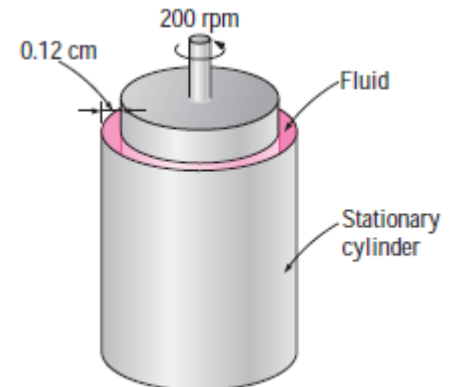


17- A lubricated shaft rotates inside a concentric sleeve bearing at 1200 rpm. The clearance δ is small compared with the radius R , so a linear velocity distribution in the lubricant may be assumed. What is the power required to rotate the shaft? $R = 2$ cm, $L = 6$ cm, $\delta = 0.1$ mm and $\mu = 0.2$ N.s/m², where L is the bearing length. (Ans: 95.3 W)

18- The clutch system shown is used to transmit torque through a 3-mm-thick oil film with $\mu = 0.38$ N.s/m² between two identical 30-cm-diameter disks. When the driving shaft rotates at a speed of 1450 rpm, the driven shaft is observed to rotate at 1398 rpm. Assuming a linear velocity profile for the oil film, determine the transmitted torque.



19- The viscosity of a fluid is to be measured by a viscometer constructed of two 75-cm-long concentric cylinders. The outer diameter of the inner cylinder is 15 cm, and the gap between the two cylinders is 0.12 cm. The inner cylinder is rotated at 200 rpm, and the torque is measured to be 0.8 N.m. Determine the viscosity of the fluid.



20- Determine the gage pressure inside a soap bubble of diameter (a) 0.2 cm and (b) 5 cm at 20°C. Take the surface tension coefficient of the soap bubble $\gamma = 0.025$ N/m.

sheet (1)

Properties of Fluids

1-Write the dimensional representation and the SI units of the following:

- | | | | |
|----------------------|------------------------|--------------------|--------------|
| a) Density | b) Specific Weight | c) Pressure | d) Energy |
| e) γh | f) Discharge | g) Power | h) ρQv |
| i) Dynamic Viscosity | j) Kinematic Viscosity | k) Surface Tension | l) $V^2/2g$ |

Solution:-

Parameter	S.I. Units	Dimensional representation
a) Density (ρ)	Kg/m^3	$M L^{-3}$
b) specific weight (γ)	$\frac{N}{m^3} = \frac{\text{Kg } m/s^2}{m^3} = \text{Kg}/m^2.s^2$	$M L^{-2} T^{-2}$
c) Pressure ($P = \frac{F}{A}$)	$P_a = \frac{N}{m^2} = \frac{\text{Kg } m/s^2}{m^2} = \text{Kg}/m.s^2$	$M L^{-1} T^{-2}$
d) Energy $\equiv F \times \underset{\substack{\uparrow \\ \text{Force}}}{\text{distance}}$	$N.m = \text{Kg } \frac{m}{s^2} m = \text{Kg } m^2/s^2$ $\equiv \text{Joule (J)}$	$M L^2 T^{-2}$
e) $\gamma h \equiv \text{Pressure}$	$P_a = \frac{N}{m^2} = \text{Kg}/m.s^2$	$M L^{-1} T^{-2}$
f) Discharge (Q) $\equiv \text{Volume flow rate}$ $\equiv \frac{V}{t}$	m^3/s	$L^3 T^{-1}$
g) Power $\equiv [\text{Energy}/\text{time}]$	$J/s = \frac{\text{Kg } m^2}{s^3} \equiv \text{watt}$	$M L^2 T^{-3}$
h) $\rho Q V \equiv m V$ $\equiv \text{Momentum Force}$	$N \equiv \frac{\text{Kg} \cdot m}{s^2}$	$M L T^{-2}$

i) Dynamic Viscosity (μ)	$\text{Pa.s} = \text{Kg/m.s}$	$\text{ML}^{-1}\text{T}^{-1}$
j) Kinematic Viscosity ($\nu = \frac{\mu}{\rho}$)	m^2/s	L^2T^{-1}
k) surface tension ($\sigma = F/L$)	$\frac{\text{N}}{\text{m}} = \frac{\text{Kg.m}}{\text{m.s}^2} = \text{Kg/s}^2$	MT^{-2}
l) $\frac{V^2}{2g} \equiv$ Velocity head	$\frac{\text{m}^2}{\text{s}^2} \div \frac{\text{s}^2}{\text{m}} = \text{m}$	L

2- If the specific weight of a gas is 12.40 N/m^3 , what is its specific volume in m^3/kg ?

Solution:-

$$\gamma = \rho \times g = 12.4$$

$$\rho = \frac{1}{\nu}$$

$$\therefore \frac{g}{\nu} = 12.4$$

$$\therefore \nu = \frac{9.81}{12.4} = 0.791 \text{ m}^3/\text{kg}$$

3 - Define:-

a) internal, external and open channel Flows

solution:-

External flow:- is the flow of an unbounded fluid over a surface such as a plate, a wire or a pipe

Internal flow:- is the flow in a pipe or duct if the fluid is completely bounded by solid surfaces

b) In compressible flow and incompressible fluid. Must the flow of a compressible fluid necessarily be treated as compressible?

solution:-

incompressible flow [$\rho \cong \text{const.}$]

A Fluid flow during which the density of the fluid remains nearly constant

incompressible fluid:-

A fluid whose density is practically independent of pressure (such as liquid)

The flow of compressible fluid (such as air) does not necessarily need to be treated as compressible fluid may still remain nearly constant during flow - especially flow at low speed

c) What is the no-slip condition? what causes it?

A fluid in direct contact with a solid surface sticks to the surface and there is no slip. This is known as the no slip condition, and it is due to the viscosity of fluid.

d) Define stress, normal stress, shear stress and pressure

stress:- is defined as force per unit area, upon which it acts

normal stress:- the normal component of a force acting on a surface per unit area

shear stress:- the tangential component of a force acting on a surface per unit area. In fluid at rest, the normal stress is called pressure

e) Can the coefficient of compressibility of fluid be negative?
How about The coefficient of Volume expansion?

solution:- The coefficient of compressibility:-

$$K = - \frac{\Delta P}{\frac{\Delta V}{V}} = \frac{\Delta P}{\frac{\Delta P}{\rho}} \quad (T = \text{const.})$$

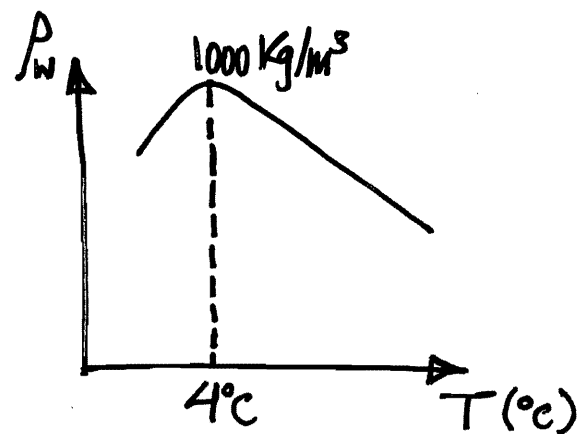
The coefficient of Volume expansion:-

$$\beta = \frac{\Delta V/V}{\Delta T} = \frac{-\Delta P/P}{\Delta T} \quad (P = \text{const.})$$

- The coefficient of compressibility of a fluid can not be negative
- but the coefficient of volume expansion can be negative (ex: liquid water below 4°C)

Note: This is reason that ice floats on water

- عندما تنخفض درجة حرارة الماء أقل من (4°C) فإن كثافة الماء تنخفض ولذلك فإن الثلج يطفو فوق سطح الماء ويظل الماء في الأسفل سائل وتظهر الريحان المائية



f) Consider two identical small glass balls dropped into two identical containers one filled with water and the other with oil. Which ball will reach the bottom of the container first? why?

The ball dropped in water will reach the bottom of the container first because of much lower viscosity of water relative to oil

4-Determine the mass and the weight of the air contained in a room whose dimensions are 6 m x 6 m x 8 m. Assume the density of the air is 1.16 kg/m³.

solution:-

Req: m_{air} , W_{air}

$$V_{air} = 6 \times 6 \times 8 = 288 \text{ m}^3$$

$$\rho_{air} = \frac{m_{air}}{V_{air}} \quad \therefore 1.16 = \frac{m_{air}}{288} \quad \therefore m_{air} = 334.08 \text{ Kg}$$

$$W_{air} = m_{air} \times g = 334.08 \times 9.81 = 3277.32 \text{ N}$$

5-Determine the specific weight, density and specific gravity of a fluid that occupies volume of 200 lit, and weighs 178 kg. Will this fluid float on the surface of an oil of specific gravity (0.80)? Provide Results in S.I. units.

$$V = 200 \text{ lit} = 200 \times 10^{-3} = 0.2 \text{ m}^3$$

$$m = 178 \text{ Kg} \quad , \quad S.G_{oil} = 0.8$$

Req: γ_f , ρ_f , $S.G_f$, will float or not with oil

solution:-

$$\gamma_f = \rho_f g = \frac{m}{V} \times g = \frac{178}{0.2} \times 9.81 = 8730.9 \text{ N/m}^3$$

$$\rho_f = \frac{m}{V} = \frac{178}{0.2} = 890 \text{ Kg/m}^3$$

$$S.G_f = \frac{\rho_f}{\rho_{H_2O @ 4^\circ C}} = \frac{890}{1000} = 0.89 > S.G_{oil} = 0.8$$

$\therefore$ Fluid will not float on oil

6-The specific gravity of ethyl alcohol is 0.79. Calculate its specific weight and its mass density. Also, The kinematic viscosity of an oil, if the density of the oil is 800 kg/m^3 and its dynamic viscosity is $6 \times 10^{-3} \text{ kg/m.s}$

Solution:-

For Ethyl alcohol: $S.G = 0.79$ $\rho_f = S.G * \rho_{H_2O} = 0.79 * 1000 = 790 \text{ Kg/m}^3$
 4°C

$$\therefore \gamma = \rho * g = 790 * 9.81 = 7749.9 \text{ N/m}^3$$

For oil: Req μ_{oil} $\rho_{oil} = 800 \text{ Kg/m}^3$ $\mu_{oil} = 0.006 \text{ Kg/ms}$

$$\mu_{oil} = \frac{\mu}{\rho_{oil}} = \frac{0.006}{800} \quad \therefore \mu_{oil} = 7.5 \times 10^{-6} \text{ m}^2/\text{s}$$

7-Water at 15°C and 1 atm pressure is heated to 100°C at constant pressure. Take the coefficient of volume expansion as $0.484 \times 10^{-3} \text{ K}^{-1}$, determine the change in the density of water. Consider the density of water at 1 atm is 999 kg/m^3 .

$$T_1 = 15^\circ\text{C} \quad , \quad T_2 = 100^\circ\text{C} \quad , \quad \beta = 0.484 \times 10^{-3} \text{ K}^{-1} \quad , \quad \rho_1 = 999 \text{ Kg/m}^3$$

Req $\Delta \rho$, ρ_2

solution:- $\beta = \frac{-(\Delta \rho / \rho)}{\Delta T}$ $0.484 \times 10^{-3} = \frac{-(\Delta \rho / 999)}{(100 - 15)}$

$$\therefore \Delta \rho = -41.1 \text{ Kg/m}^3 \quad \therefore \rho_2 - \rho_1 = -41.1 = \Delta \rho$$

$$\therefore \rho_2 = \rho_1 + \Delta \rho = 999 - 41.1 = 957.9 \text{ Kg/m}^3$$

8-A water tank is completely filled with liquid water at 20°C. The tank material is such that it can withstand tension caused by a volume expansion of 2 percent.

Determine the maximum temperature rise allowed without jeopardizing safety. Take the coefficient of volume expansion as $0.377 \times 10^{-3} \text{ K}^{-1}$.

$$T_i = 20^\circ\text{C} \quad , \quad \frac{\Delta V}{V} = 0.02 \quad , \quad \beta = 0.377 \times 10^{-3} \text{ K}^{-1} \quad \underline{\text{Req}} \quad \Delta T_{\max}$$

Solution:- Volume expansion of 2% $\Rightarrow \left[\frac{\Delta V}{V} = 0.02 \right]$

$$\therefore \frac{V_f - V_i}{V_i} = 0.02 \quad \rightarrow \quad \therefore V_f = 1.02 V_i$$

$$\therefore \beta = \frac{(\Delta V/V)}{\Delta T_{\max}} \quad \therefore 0.377 \times 10^{-3} = \frac{0.02}{\Delta T_{\max}}$$

$$\therefore \Delta T_{\max} = 53.05^\circ\text{C}$$

9-In a fluid the velocity measured at a distance of 75mm from the boundary is 1.125m/s. The fluid has absolute viscosity 0.048 Pa s and relative density 0.913. What is the velocity gradient and shear stress at the boundary assuming a linear velocity distribution?

$$\mu = 0.048 \text{ Pa.s} \quad , \quad \rho_r = \text{S.G} = 0.913 \quad \therefore \rho_{\text{fluid}} = 0.913 \times 1000 = 913 \text{ Kg/m}^3$$

Req:- Velocity gradient $= \frac{dV}{dy}$ $\sim \gamma$ [Assuming a linear velocity distribution]

Solution:-

$$\frac{dV}{dy} = \frac{\Delta V}{\Delta y} = \frac{1.125 - 0}{0.075} = 15 \text{ s}^{-1}$$

$$\tau = \mu \frac{dV}{dy} = \mu \frac{\Delta V}{\Delta y} = 0.048 \times 15 = 0.72 \text{ Pa}$$

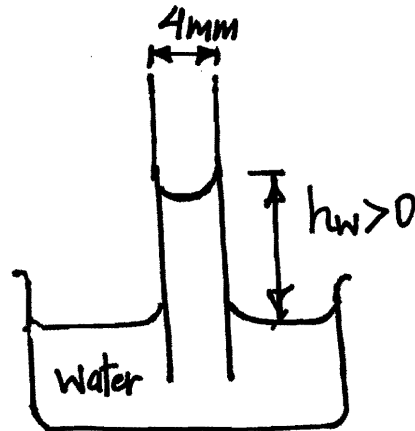
10- Calculate the capillary effect in millimeters in a glass tube of 4 mm diameter., when immersed in (i) water and (ii) in mercury. The temperature of liquid is 20° C and the values of surface tension of water and mercury at this temperature in contact with air are 0.0075 kg/m and 0.052 kg/m. The contact angle for water = 0 and for mercury = 130°.

i- Water

$$\sigma_w = 0.0075 \text{ (Kg/m)} \quad \theta_w = 0^\circ$$

$$h_{w, \text{rise}} = \frac{2 \sigma_w \cos \theta_w}{\gamma_w R}$$

$$= \frac{2 \times (0.0075 \times 9.81) \cos 0}{9810 \times 2 \times 10^{-3}}$$



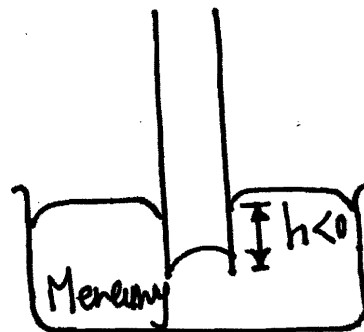
$$h_{w, \text{rise}} = 7.5 \times 10^{-3} \text{ m} = 7.5 \text{ mm}$$

ii- Mercury

$$\sigma_{Hg} = 0.052 \text{ Kg/m} \quad \theta_{Hg} = 130^\circ$$

$$h_m = \frac{2 \sigma_m \cos \theta_m}{\gamma_m R}$$

$$= \frac{2 \times (0.052 \times 9.81) \cos 130}{13600 \times 9.81 \times 2 \times 10^{-3}}$$



$$\therefore h_m = -2.46$$

$$h_{m, \text{drop}} = -2.46 \text{ mm}$$

-ve sign $\Rightarrow$ indicates capillary drop

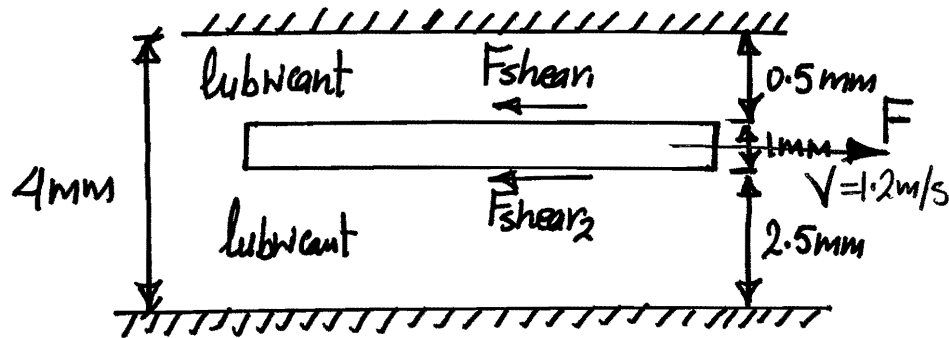
11-Determine the force required to pull a 1.0 mm thick plate between two stationary parallel plates separated by a gap 4 mm. the moving plate is $(2 \times 3) \text{ m}^2$ surface area, and is located 0.5 mm from one of stationary plates. The moving plate has a velocity of 1.2 m/s and the viscosity of the lubricant is 0.07 poise.

$$A = 2 \times 3 = 6 \text{ m}^2$$

$$\mu_{\text{lubricant}} = 0.07 \text{ Poise}$$

$$= 0.007 \text{ Poise}$$

Req: F
solution:-



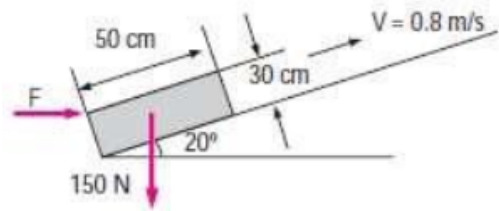
$$F_{\text{shear}} = \mu \frac{dV}{dy} A$$

$$F = F_{\text{shear}_1} + F_{\text{shear}_2} = \tau_1 A + \tau_2 A = A(\tau_1 + \tau_2)$$

$$= \left(\mu \frac{dV}{dy}_1 + \mu \frac{dV}{dy}_2 \right) A = \mu A \left(\frac{\Delta V_1}{\Delta y_1} + \frac{\Delta V_2}{\Delta y_2} \right)$$

$$= (0.007 \times 6) \times \left(\frac{1.2 - 0}{0.5 \times 10^{-3}} + \frac{1.2 - 0}{2.5 \times 10^{-3}} \right) = 120.96 \text{ N}$$

12- A 50-cm x 30-cm x 20-cm block weighing 150 N is to be moved at a constant velocity of 0.8 m/s on an inclined surface with a friction coefficient of 0.27. (a) Determine the force F that needs to be applied in the horizontal direction. (b) If a 0.4-mm-thick oil film with a dynamic viscosity of 0.012 Pa.s is applied between the block and inclined surface, determine the percent reduction in the required force.



$$V = 0.8 \text{ m/s}$$

$$\mu = 0.27$$

$$\text{Req.} - F_a$$

Solution:-

Case (a):-

Dry friction
[No oil film exists]

$$\sum F_x = 0$$

$$F_a \cos 20^\circ - f F_N - 150 \sin 20^\circ = 0$$

$$\boxed{F_a \cos 20^\circ - 0.27 F_N = 150 \sin 20^\circ} \quad \text{--- ①}$$

$$\sum F_y = 0$$

$$F_N - F_a \sin 20^\circ - 150 \cos 20^\circ = 0$$

$$\boxed{-F_a \sin 20^\circ + F_N = 150 \cos 20^\circ} \quad \text{--- ②}$$

$$\boxed{F_N = 150 \cos 20^\circ + F_a \sin 20^\circ} \quad \text{--- ②'}$$

From eq ① & ②'

$$\therefore F_a \cos 20 - 0.27(150 \cos 20 + F_a \sin 20) = 150 \sin 20$$

$$\therefore F_a = 105.45 \text{ N}$$

Case (b):- [oil film exists]

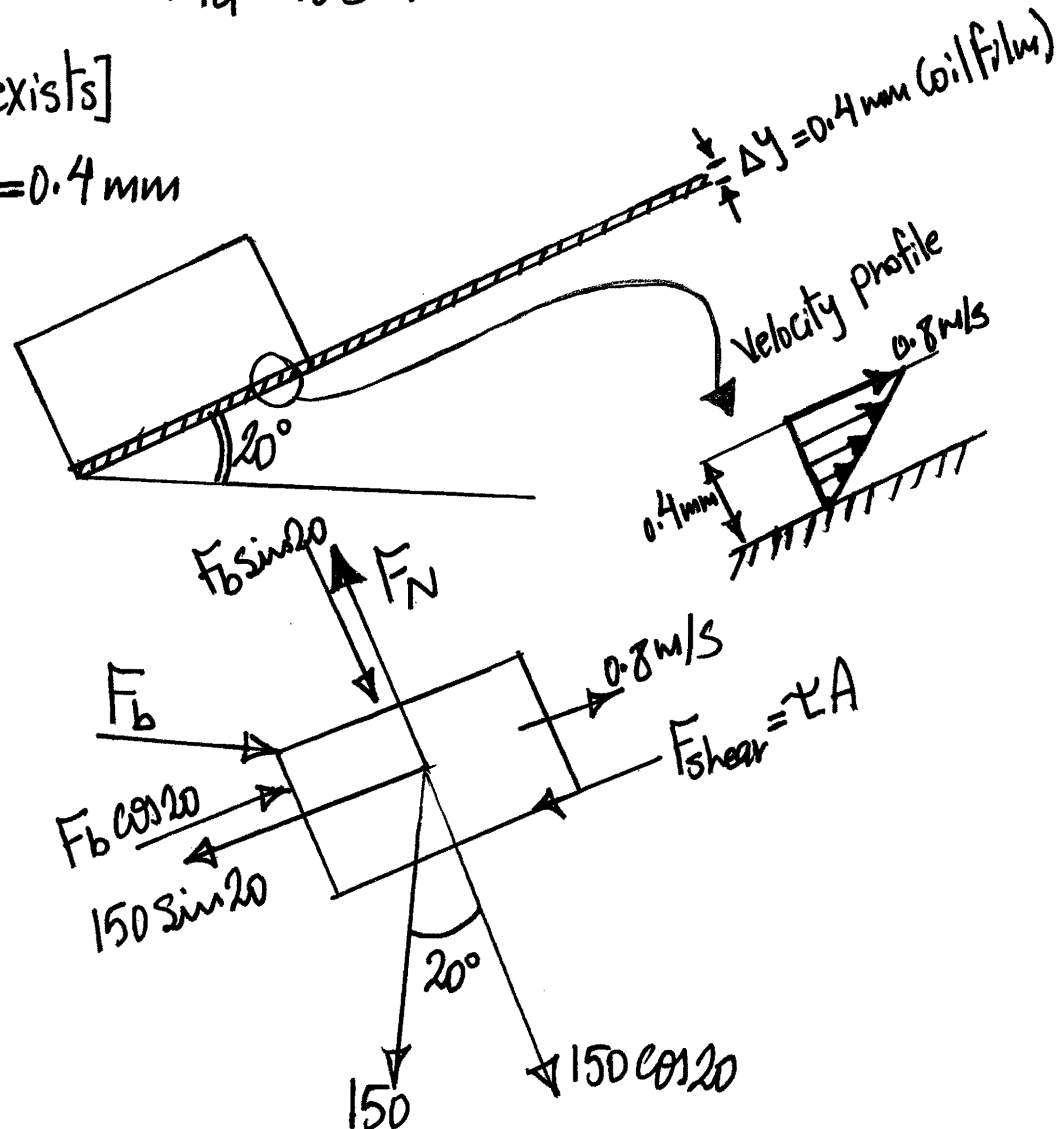
$\Delta y \equiv$ oil film thickness = 0.4 mm

$$\mu_{\text{oil}} = 0.012 \text{ Pa.s}$$

Req.: % $\Delta F_{\text{reduction}}$

Solution:

$$\sum F_x = 0$$



$$\therefore F_b \cos 20 = F_{\text{shear}} + 150 \sin 20 \quad \text{--- (1)}$$

$$F_{\text{shear}} = \tau A = \mu \frac{\Delta v}{\Delta y} A = 0.012 * (50 * 20) * 10^{-4} * \frac{(0.8 - 0)}{0.4 * 10^{-3}} = 2.4 \text{ N}$$

From eq (1)

$$\therefore F_b \cos 20 = 2.4 + 150 \sin 20$$

$$\therefore F_b = 57.15 \text{ N}$$

$$\therefore \% \Delta F_{\text{reduction}} = \frac{F_a - F_b}{F_a} * 100 = \left(\frac{105.45 - 57.15}{105.45} \right) * 100 = 45.8\%$$

13-At the ocean surface under normal atmospheric pressure, the density of sea water is 1047 kg/m^3 . The absolute pressure is about 35 bar at the deepest point in the Pacific ocean where the depth is about 10860 m. What is the percentage change in the density of the sea water between the surface and the ocean bottom? Bulk modulus of elasticity of sea water = 2338.67 MPa.

$$\frac{\Delta \rho}{\rho} \% \quad E = 2338.67 \text{ MPa}$$

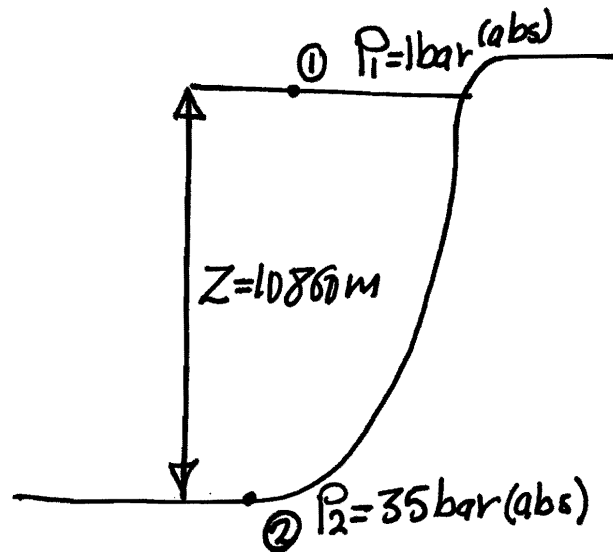
sea water

$$K = \frac{dP}{\frac{d\rho}{\rho}}$$

$$2338.67 \times 10^6 = \frac{(35-1) \times 10^5}{\frac{d\rho}{\rho}}$$

$$\therefore \frac{d\rho}{\rho} = 0.00145$$

$$= \frac{d\rho}{\rho} = 0.145 \%$$



Exact solution

$$dP = \rho g dz$$

$$\therefore K = \frac{dP}{\frac{d\rho}{\rho}} = \rho \frac{dP}{d\rho} = \rho^2 g \frac{dz}{d\rho}$$

$$= K = \rho^2 g \frac{dz}{d\rho}$$

$$\therefore \int_{\rho_1}^{\rho} \frac{d\rho}{\rho^2} = \int_0^z \frac{g}{K} dz$$

$$\therefore -\frac{1}{\rho} \Big|_{\rho_1}^{\rho} = \frac{g}{K} z$$

$$\therefore -\frac{1}{\rho} + \frac{1}{\rho_1} = \frac{g}{K} z$$

$$\therefore \rho = \frac{1}{\left(\frac{1}{\rho_1} - \frac{g}{K} z\right)}$$

$$\therefore \rho_2 = \frac{1}{\left(\frac{1}{1047} - 9.81 \times \frac{10860}{2338.67 \times 10^6}\right)} = 1099.4 \text{ kg/m}^3$$

$$\therefore \frac{P_2 - P_1}{P_1} = \frac{1099.4 - 1047}{1047} \times 100 = 5\%$$

14-A 10-kg block slides down a smooth inclined surface, 20° with horizontal. Determine the terminal velocity of the block if the 0.1mm gap between the block and the surface contains oil of viscosity 3.8 cP (centipoises). Assume velocity distribution in the gap is linear and the area of block in contact with the oil is 0.1 m^2 .

$m = 10 \text{ kg}$ • Linear Velocity distribution in oil gap

$$A_{\text{contact}} = 0.1 \text{ m}^2$$

$$\mu_{\text{oil}} = 3.8 \text{ cP} = 3.8 \times 10^{-3} \text{ Pa}\cdot\text{s}$$

Req V_{block}

solution:-

$$\sum F_x = 0$$

$$\therefore F_{\text{shear}} = mg \sin 20$$

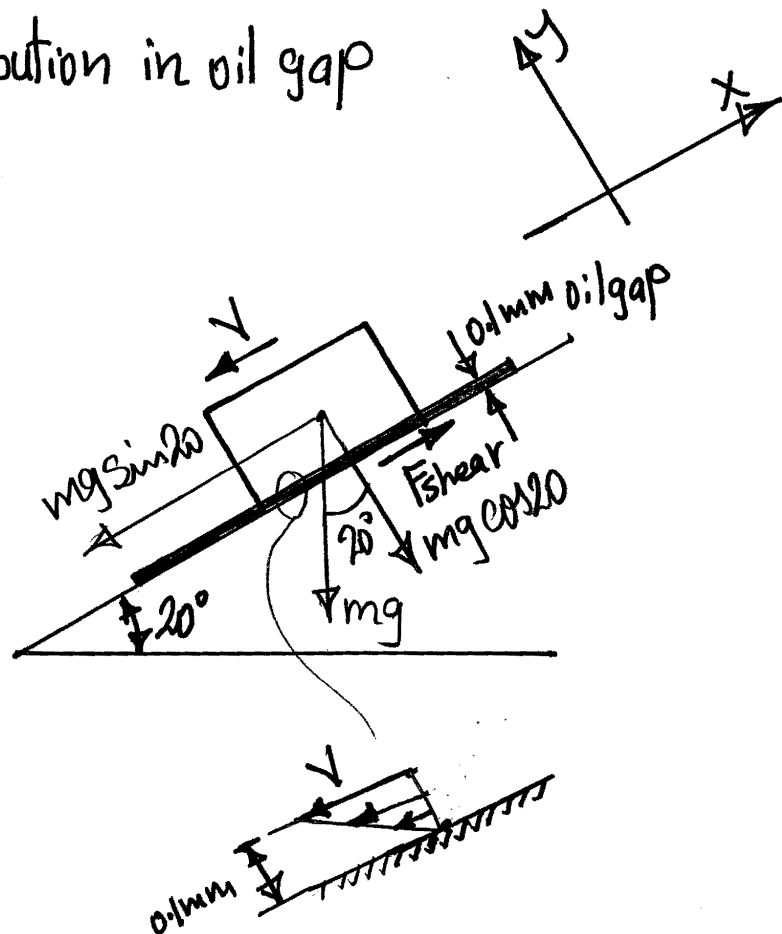
$$\tau A = mg \sin 20$$

$$\tau = \mu \frac{\Delta V}{\Delta y}$$

$$\therefore \mu \frac{\Delta V}{\Delta y} A = mg \sin 20$$

$$3.8 \times 10^{-3} \times \frac{(V-0)}{0.1 \times 10^{-3}} \times 0.1 = 10 \times 9.81 \times \sin 20$$

$$\therefore V = 8.8 \text{ m/s}$$



15-A large movable plate is located between two fixed plates at a distance 3.0 mm from one plate and 6.0 mm from the other. The spaces between the movable plate and the fixed plates are filled with oil of viscosity 0.02 and 0.01 Pa.s respectively. Determine the force required to pull the plates at 4 m/s if the surface area of the plate is 0.9 m^2 and its thickness is negligible. If one oil is used to fill the spaces between the plates ($\mu = 0.01 \text{ Pa.s}$), what is the minimum force required to pull the plate at the same velocity? Determine the location of the movable plate with respect to the fixed plates in this case.

$$A = 0.9 \text{ m}^2$$

Req: F_1 For case (1)

Solution:

$$F_1 = F_{\text{shear}_1} + F_{\text{shear}_2}$$

$$F_1 = \tau_1 A + \tau_2 A$$

$$= A \left[\mu_1 \frac{\Delta V_1}{\Delta y_1} + \mu_2 \frac{\Delta V_2}{\Delta y_2} \right]$$

$$= 0.9 \times \left[(0.02 \times \left(\frac{4-0}{3 \times 10^{-3}} \right)) + (0.01 \times \left(\frac{4-0}{6 \times 10^{-3}} \right)) \right] = 30 \text{ N}$$

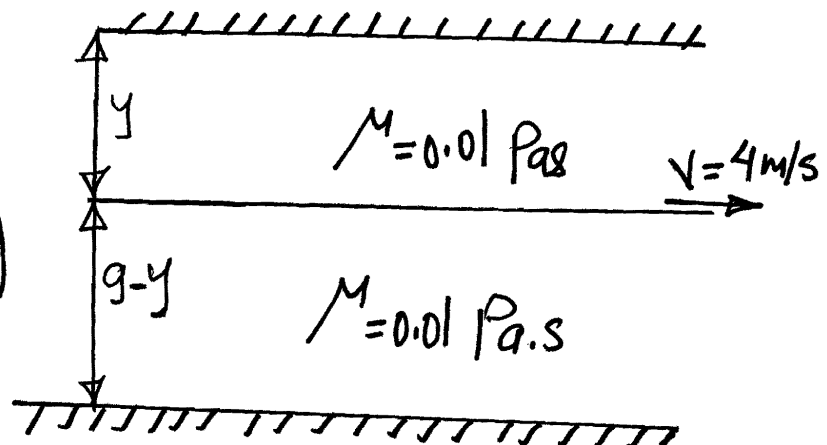
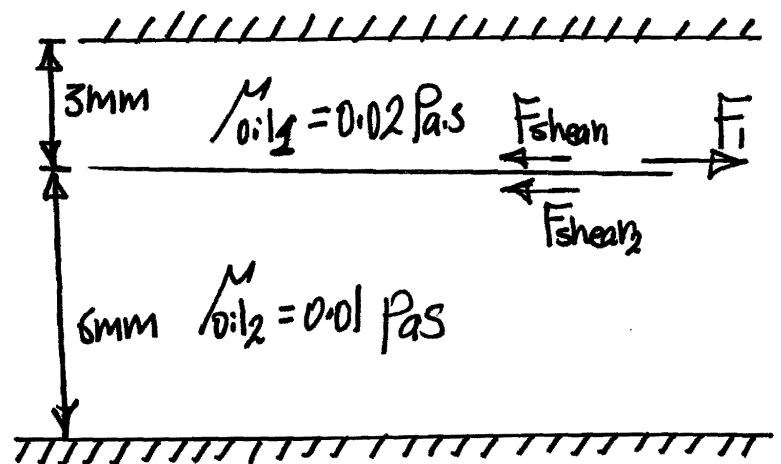
For case (2): only one oil exists on both sides of thin flat plate

Req F_{min} y

$$F = A \mu \Delta V \times \left(\frac{1}{y} + \frac{1}{(g-y)} \right)$$

$$= 0.9 \times 0.01 \times (4-0) \times \left(\frac{1}{y} + \frac{1}{(g-y)} \right)$$

$$= 0.036 \left(\frac{1}{y} + \frac{1}{g-y} \right)$$



$$\therefore \frac{dF}{dy} = 0.036 * \left[\frac{-1}{y^2} + \frac{1}{(9-y)^2} \right] = 0$$

$$\therefore \frac{1}{y^2} = \frac{1}{81 - 18y + y^2}$$

$$\therefore \cancel{y^2} = 81 - 18y + \cancel{y^2}$$

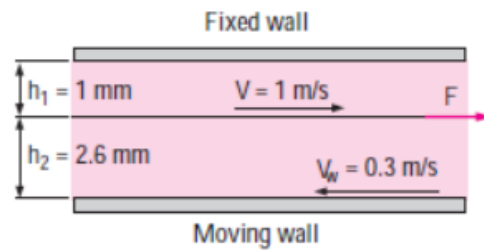
$$\therefore y = \frac{81}{18} \text{ mm} = 4.5 \text{ mm}$$

So, For $F_{\min}$, Plate should be placed in the middle

$$F_{\min} = 0.036 * \left[\frac{1}{(4.5 * 10^{-3})} + \frac{1}{(9 - 4.5) * 10^{-3}} \right]$$

$$= 16 \text{ N}$$

16- A thin 20-cm x 20-cm flat plate is pulled at 1 m/s horizontally through a 3.6-mm-thick oil layer sandwiched between two plates, one stationary and the other moving at a constant velocity of 0.3 m/s. The dynamic viscosity of oil is 0.027 Pa.s. Assuming the velocity in each oil layer to vary linearly, (a) plot the velocity profile and find the location where the oil velocity is zero and (b) determine the force that needs to be applied on the plate to maintain this motion.



$$A = (20 \times 20) \text{ cm}^2 = 400 \text{ cm}^2$$

$\mu = 0.027 \text{ Pa.s}$: Linear Velocity Variation in each oil Film

Req: y (location where oil velocity = 0)

solution:

$$\frac{y}{2.6 - y} = \frac{0.3}{1}$$

$$\therefore y = 0.78 - 0.3y$$

$$\therefore y = 0.6 \text{ mm}$$

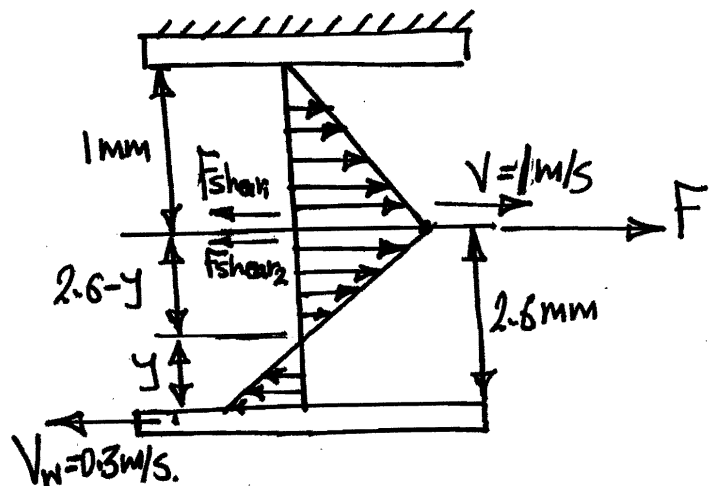
$$F = F_{\text{shear}_1} + F_{\text{shear}_2}$$

$$= \left(\mu \frac{\Delta V_1}{\Delta y_1} + \mu \frac{\Delta V_2}{\Delta y_2} \right) A = A \mu \left(\frac{\Delta V_1}{\Delta y_1} + \frac{\Delta V_2}{\Delta y_2} \right)$$

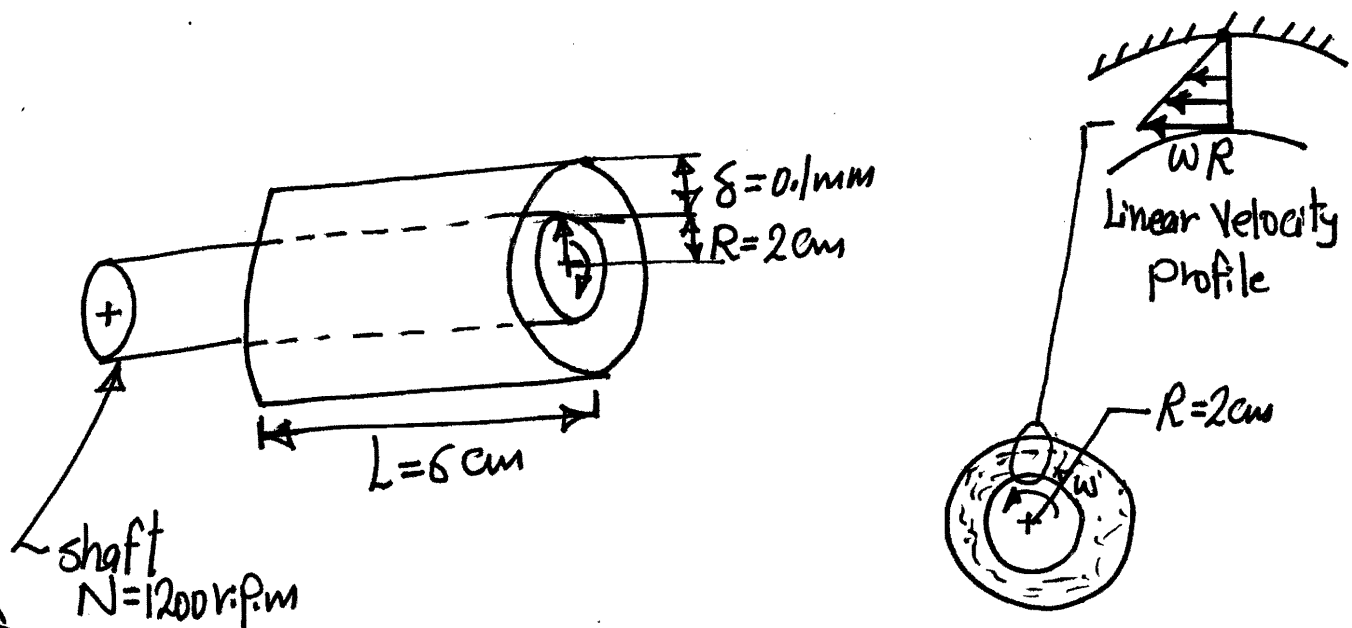
$$\Delta V_1 = (1 - 0) = 1 \text{ m/s}, \Delta y_1 = 1 \text{ mm}$$

$$\Delta V_2 = 1 - (-0.3) = 1.3 \text{ m/s}, \Delta y_2 = 2.6 \text{ mm}$$

$$F = A \mu \left(\frac{\Delta V_1}{\Delta y_1} + \frac{\Delta V_2}{\Delta y_2} \right) = 400 \times 10^{-4} \times 0.027 \times \left(\frac{1}{1 \times 10^{-3}} + \frac{1.3}{2.6 \times 10^{-3}} \right) = 1.62 \text{ N}$$



17-A lubricated shaft rotates inside a concentric sleeve bearing at 1200 rpm. The clearance δ is small compared with the radius R , so a linear velocity distribution in the lubricant may be assumed. What is the power required to rotate the shaft? $R = 2 \text{ cm}$, $L = 6 \text{ cm}$, $\delta = 0.1 \text{ mm}$ and $\mu = 0.2 \text{ N.s/m}^2$, where L is the bearing length. (Ans: 95.3 W)



Req $\hat{P}_{\text{shaft}}$

solution: $\hat{P}_{\text{shaft}} = T * \omega = (F_{\text{shear}} * R) * \omega = \tau A * R * \omega$

$$= \left[\mu \frac{\Delta V}{\delta} \right] * A * R * \omega$$

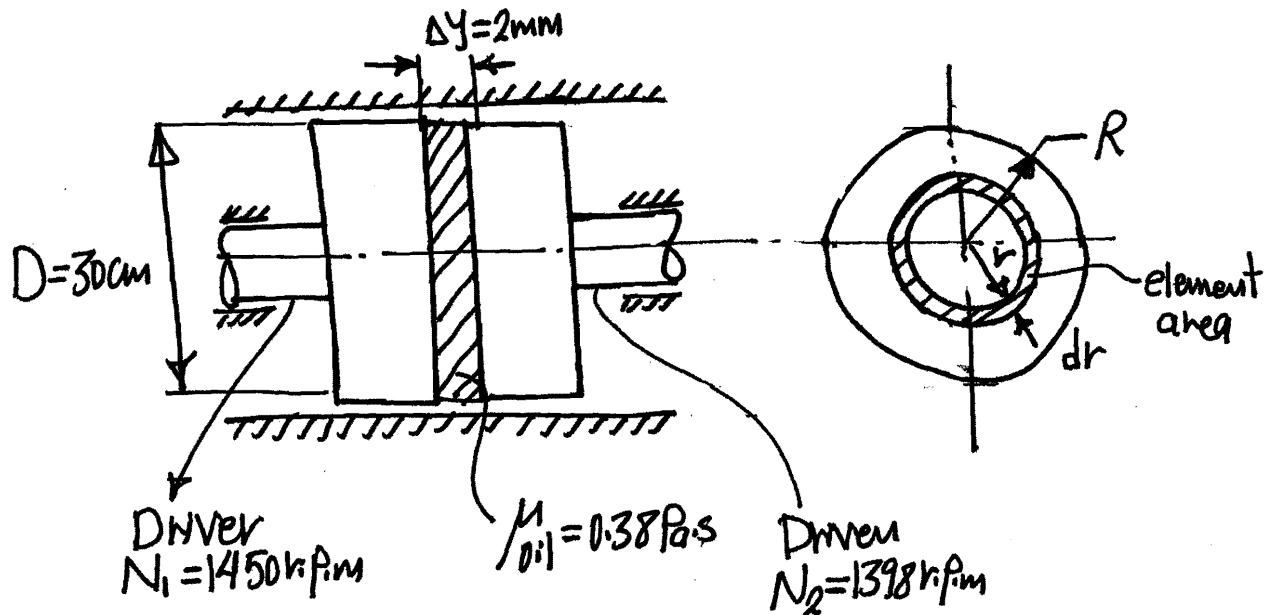
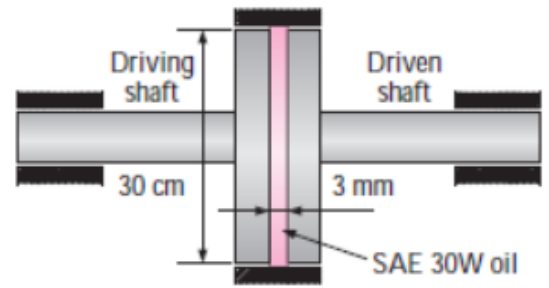
$$= \mu \frac{(\omega R - 0)}{\delta} (2\pi R L) * R * \omega$$

$$= \frac{2\pi \mu L \omega^2}{\delta} R^3 = \frac{2\pi \mu L \left(\frac{2\pi N}{60} \right)^2}{\delta} R^3$$

$$= \frac{2\pi * 0.2 * 0.06 * \left(\frac{2\pi * 1200}{60} \right)^2}{(0.0001)} * (0.02)^3$$

$$= 95.25 \text{ W}$$

18- The clutch system shown is used to transmit torque through a 3-mm-thick oil film with $\mu = 0.38 \text{ N} \cdot \text{s/m}^2$ between two identical 30-cm-diameter disks. When the driving shaft rotates at a speed of 1450 rpm, the driven shaft is observed to rotate at 1398 rpm. Assuming a linear velocity profile for the oil film, determine the transmitted torque.



For element area "dA", is:-

$$dA = 2\pi r dr$$

- Torque element area "dA" :-

$$dT = dF_{\text{shear}} r$$

$dF_{\text{shear}} \equiv$ shear force on differential area "dA"

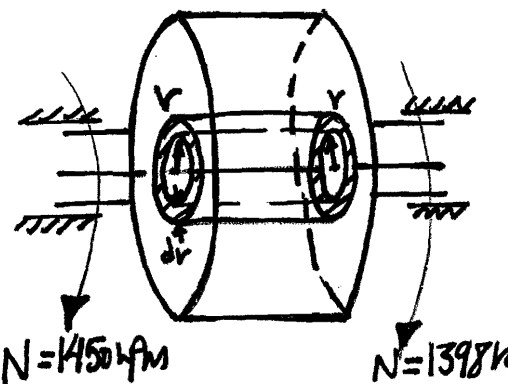
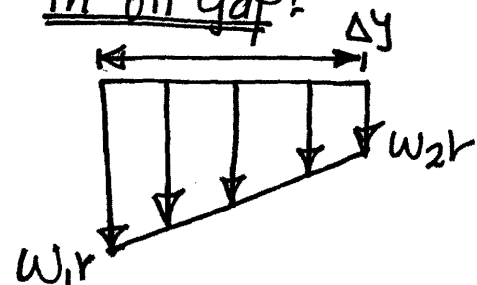
$$dF_{\text{shear}} = \tau dA = \mu \frac{\Delta V}{\Delta y} 2\pi r dr$$

$$\therefore dT = \mu \frac{\Delta V}{\Delta y} 2\pi r^2 dr$$

$$= \mu \frac{\omega_1 r - \omega_2 r}{\Delta y} (2\pi r^2) dr$$

$$= \mu \frac{\omega_1 - \omega_2}{\Delta y} 2\pi r^3 dr = \frac{2\pi \mu (\omega_1 - \omega_2)}{\Delta y} r^3 dr$$

Linear Velocity profile in oil gap:-



$$\therefore T = \int dT = \frac{2\pi \mu (w_1 - w_2)}{\Delta y} \int_0^R r^3 dr$$

$$= \frac{2\pi \mu (w_1 - w_2)}{\Delta y} \frac{R^4}{4}$$

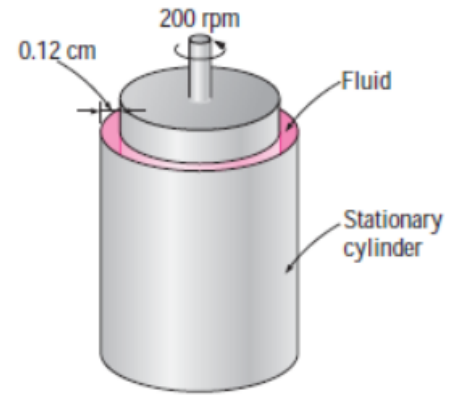
$$= \frac{\pi \mu (w_1 - w_2)}{\Delta y} \frac{R^4}{2} = \frac{\pi \mu (w_1 - w_2)}{\Delta y} \frac{D^4}{32}$$

$$\therefore T = \frac{\pi \mu (w_1 - w_2) D^4}{32 \Delta y}$$

$$= \frac{\pi \times 0.38 \times \frac{2\pi}{60} (1450 - 1398) \times (0.3)^4}{32 \times 0.002}$$

$$= 0.823 \text{ N.m}$$

19- The viscosity of a fluid is to be measured by a viscometer constructed of two 75-cm-long concentric cylinders. The outer diameter of the inner cylinder is 15 cm, and the gap between the two cylinders is 0.12 cm. The inner cylinder is rotated at 200 rpm, and the torque is measured to be 0.8 N.m. Determine the viscosity of the fluid.



$$R = 7.5 \text{ cm}, \delta = 0.12 \text{ cm},$$

$$N = 200 \text{ r.p.m.}, T_{\text{meas}} = 0.8 \text{ N.m}$$

$$\underline{\text{Req}} \mu, L = 75 \text{ cm}$$

$$T = F \cdot R = \tau A R$$

$$= \mu \frac{\Delta V}{\delta} (2\pi R) L R$$

$$= \mu \frac{(\omega R - 0)}{\delta} (2\pi R) L R$$

$$= \frac{2\pi \mu L \omega}{\delta} R^3$$

$$\therefore \mu = \frac{\delta \cdot T}{2\pi L \left(\frac{2\pi N}{60}\right) R^3} = \frac{0.0012 \cdot 0.8}{2\pi \cdot 0.75 \cdot \left(\frac{2\pi \cdot 200}{60}\right) \cdot (0.075)^3}$$

$$= 0.0231 \text{ Pa.s}$$

20- Determine the gage pressure inside a soap bubble of diameter (a) 0.2 cm and (b) 5 cm at 20°C. Take the surface tension coefficient of the soap bubble $\gamma = 0.025 \text{ N/m}$.

$$\Delta P_{\text{bubble}} = P_i - P_o = \frac{4\gamma}{R}, P_o = P_{\text{atm}} = 0 \text{ (gage)} \underline{\text{Req}}: P_{i, \text{gage } a, b}$$

$$\underline{\text{Case (a)}}: R_a = 0.1 \text{ cm} \therefore P_{i, \text{gage } a} = \left(\frac{4 \cdot 0.025}{0.001}\right) = 100 \text{ Pa}$$

$$\underline{\text{Case (b)}}: R_b = 2.5 \text{ cm} \therefore P_{i, \text{gage } b} = \left(\frac{4 \cdot 0.025}{2.5 \times 10^{-2}}\right) = 4 \text{ Pa}$$